

5

Visible and Ultraviolet Molecular Spectroscopy

5.1. INTRODUCTION

Probably the first physical method used in analytical chemistry was based on the quality of the color in colored solutions. The first things we observe regarding colored solutions are their *hue*, or color, and the color's *depth*, or *intensity*. These observations led to the technique historically called *colorimetry*; the color of a solution could identify species (qualitative analysis) while the intensity of the color could identify the concentration of the species present (quantitative analysis). This technique was the first use of what we now understand to be absorption spectroscopy for chemical analysis. When white light passes through a solution and emerges as red light, we say that the solution is red. What has actually happened is that the solution has allowed the red component of white light to pass through, whereas it has absorbed the complementary colors, yellow and blue. The more concentrated the sample solution, the more yellow and blue light is absorbed and the more intensely red the solution appears to the eye. For a long time, experimental work made use of the human eye as the detector to measure the hue and intensity of colors in solutions. However, even the best analyst can have difficulty comparing the intensity of two colors with slightly different hues, and there are of course people who are color-blind and cannot see certain colors. Instruments have been developed to perform these measurements more accurately and reliably than the human eye. While the human eye can only detect visible light, this chapter will focus on both the ultraviolet (UV) and the visible (VIS) portions of the spectrum.

The wavelength range of UV radiation starts at the blue end of visible light (about 400 nm) and ends at approximately 200 nm for spectrometers operated in air. The radiation has sufficient energy to excite valence electrons in many atoms and molecules; consequently, UV radiation is involved with electronic excitation. Visible light, considered to be light with wavelengths from 800 to 400 nm, acts in the same way as UV light. It is also considered part of the electronic excitation region. For this reason we find commercial spectroscopic instrumentation often operates with wavelengths between 800 and 200 nm. Spectrometers of this type are called UV/Visible (or UV/VIS) spectrometers. The vacuum UV region of the spectrum extends below 200 nm to the X-ray region of the spectrum, at $\sim 100 \text{ \AA}$. It is called the vacuum UV region because oxygen, water vapor, and other molecules in air absorb UV radiation below 200 nm, so the spectrometer light path must be free of air to observe wavelengths $< 200 \text{ nm}$. The instrument must be evacuated (kept under vacuum) or purged with an appropriate non-UV absorbing gas such as helium for this region to be used. Vacuum UV radiation is also involved in

electronic excitation but the spectrometers are specialized and not commonly found in undergraduate or routine analytical laboratories. For our purposes the term UV will mean radiation between 200 and 400 nm, unless stated otherwise. The major types of analytical spectroscopy operating within this wavelength range are listed in Table 5.1. This chapter will focus on molecular spectroscopy—the absorption and emission of UV and visible radiation by molecules and polyatomic species. We will also look at the use of scattering of visible light to provide information about macromolecules and particles. AAS is covered in Chapter 6 and atomic emission spectroscopy in Chapter 7.

The interaction of UV and visible radiation with matter can provide qualitative identification of molecules and polyatomic species, including ions and complexes. Structural information about molecules and polyatomic species, especially organic molecules, can be acquired. This qualitative information is usually obtained by observing the UV/VIS **spectrum**, the absorption of UV and visible radiation as a function of wavelength by molecules. A typical UV absorption spectrum is shown in Fig. 5.1. The spectrum may be plotted as wavelength vs. absorbance, transmittance, or molar absorptivity, ϵ . The molar absorptivity is defined subsequently. In Fig. 5.1, the absorption spectrum of pyridine dissolved in ethanol is plotted as $\log \epsilon$ vs. wavelength in ångströms (Å).

Quantitative information can also be obtained by studying the absorption or emission of UV and visible radiation by molecules or polyatomic species. As a very simple example, we can look at the absorption spectrum of a red solution such as red ink in water (Fig. 5.2). It can be seen that with a colorless sample of pure water, shown as the dotted line, all wavelengths of white light, including all of the red wavelengths, are transmitted through the sample. If we add one drop of red ink to water, to make a solution that appears pale red, the spectrum shows that some of the blue and some of the yellow light

Table 5.1 Spectroscopy Using UV and Visible Light

Function	Analytical field	Analytical application
Atomic Spectroscopy		
Absorption of UV/VIS radiation	Atomic absorption spectrometry	Quantitative elemental analysis
Emission of UV/VIS radiation	Flame photometry, atomic emission spectrometry	Qualitative and quantitative multielemental analysis
Emission of UV/VIS radiation	Atomic fluorescence spectrometry	Quantitative elemental analysis of ultratrace concentrations (sub-ppb)
Molecular Spectroscopy		
Absorption of UV/VIS radiation	UV/VIS Molecular absorption spectroscopy, spectrophotometry	Qualitative and quantitative determinations of aromatic and unsaturated organic compounds, including natural products; direct and indirect quantitative determination of inorganic ions, organic molecules, and biochemicals
Emission of UV/VIS radiation	Molecular fluorescence, Molecular phosphorescence	Detection of small quantities (<ng) of certain aromatic compounds and natural products; analysis of gels and glasses; determination of organic and inorganic species by “tagging”

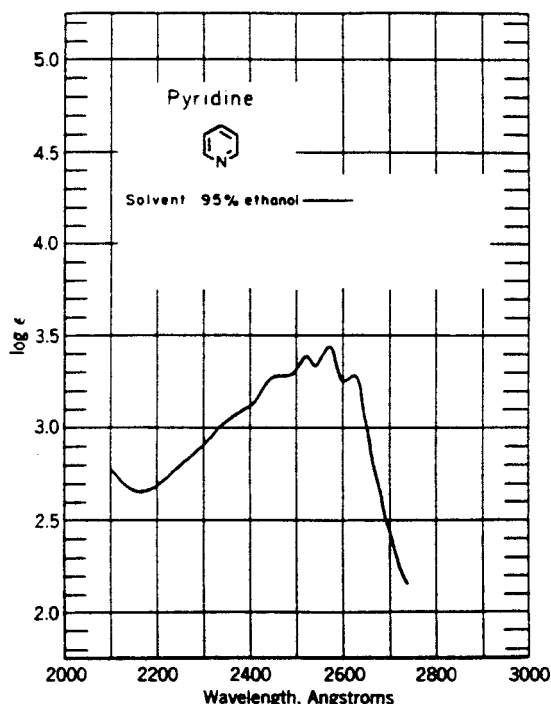


Figure 5.1 A typical UV/VIS absorption spectrum for an organic molecule in solution. The spectrum is that of pyridine dissolved in 95% ethanol. The absorption wavelength is plotted on the x -axis and the logarithm of the molar absorptivity is plotted on the y -axis. (From Jaffé and Orchin. Reprinted with the permission of Professor M. Orchin.)

have been absorbed, but all of the red light has been transmitted. If we add more red ink to the water to make a dark red solution, most of the blue and yellow light has been absorbed, but again all of the red light has been transmitted. The amount of red light falling on the eye or the detector is the same in each case; the amount of ink in the solution is related to the blue and yellow light absorbed, not to the color transmitted. We could construct a series of known amounts of red ink in water and quantitatively measure other ink solutions by measuring the amount of light absorbed at, for example, 450 nm. Concentrations of species in samples, especially solutions, are often measured using UV/VIS absorption spectrometry or fluorescence spectrometry. The measurement of concentrations or changes in concentrations can be used to calculate equilibrium constants, reaction kinetics, and stoichiometry for chemical systems. Quantitative measurements by UV/VIS spectrometry are important in environmental monitoring, industrial process control, pharmaceutical quality control, and clinical chemistry, to name a just a few areas. The emission of radiation by molecules may occur in several ways following excitation of the molecule; two processes are fluorescence and phosphorescence. These processes will be discussed in Sections 5.8–5.10.

5.1.1. Electronic Excitation in Molecules

Molecules are composed of atoms that are held together by sharing electrons to form chemical bonds. Electrons in molecules move in molecular orbitals at discrete energy

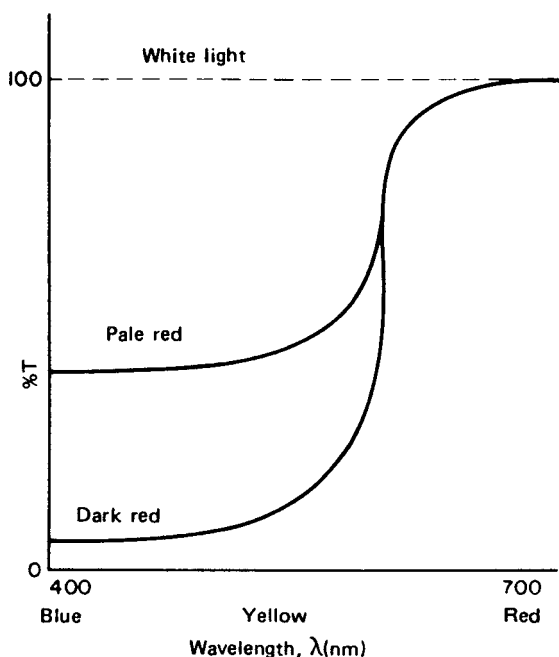


Figure 5.2 Visible absorption spectra of a colorless sample of pure water (dotted line), a pale red solution of red ink in water, and a dark red solution of red ink in water. Note that none of the samples absorbs red light.

levels as defined by quantum theory. When the energy of the electrons is at a minimum, the molecules are in the lowest energy state, or ground state. The molecules can absorb radiation and move to a higher energy state, or *excited state*. When the molecule becomes excited, an outer shell (valence) electron moves to an orbital of higher energy. The process of moving electrons to higher energy states is called *electronic excitation*. For radiation to cause electronic excitation, it must be in the visible or UV region of the electromagnetic spectrum.

The frequency absorbed or emitted by a molecule and the energy of radiation are related by $\Delta E = h\nu$. The actual amount of energy required depends on the difference in energy between the ground state E_0 and the excited state E_1 of the electrons. The relationship is described by

$$\Delta E = E_1 - E_0 = h\nu \quad (5.1)$$

where E_1 is the energy of the excited state and E_0 is the energy of the ground state.

You may want to review the topics of bonding, molecular orbitals, Lewis structures, and organic chemistry in your general chemistry textbook or in the texts by Chang or Zumdahl listed in the bibliography to help you understand the material discussed subsequently. The discussion will focus on organic molecules, as the bonding is relatively easy to understand. Inorganic molecules also undergo absorption and emission of UV and visible radiation, as do complexes of organic molecules with metal ions, but the bonding in inorganic molecules and complexes of the transition metals and heavier elements is complicated due to electrons in the d and f orbitals.

Three distinct types of electrons are involved in valence electron transitions in molecules. First are the electrons involved in single bonds, such as those between carbon and hydrogen in alkanes. These bonds are called *sigma* (σ) *bonds*. The amount of energy required to excite electrons in σ bonds is usually more than UV photons of wavelengths >200 nm possess. For this reason, alkanes and other saturated compounds (compounds with only single bonds) do not absorb UV radiation and are therefore frequently very useful as transparent solvents for the study of other molecules. An example of such a nonabsorbing compound is the alkane hexane, C_6H_{14} .

Next we have the electrons involved in double and triple (unsaturated) bonds. These bonds involve a *pi* (π) *bond*. Typical examples of compounds with π bonds are alkenes, alkynes, conjugated olefins, and aromatic compounds (Fig. 5.3). Electrons in π bonds are excited relatively easily; these compounds commonly absorb in the UV or visible region.

Electrons that are not involved in bonding between atoms are the third type of electrons in molecules. These are called *n* electrons, for nonbonding electrons. In saturated hydrocarbons the outer shell electrons of carbon and hydrogen are all involved in bonding; hence these compounds do not have any *n* electrons. Organic compounds containing nitrogen, oxygen, sulfur, or halogens, however, frequently contain electrons that are nonbonding (Fig. 5.4). Because *n* electrons are usually excited by UV or visible radiation, many compounds that contain *n* electrons absorb UV/VIS radiation.

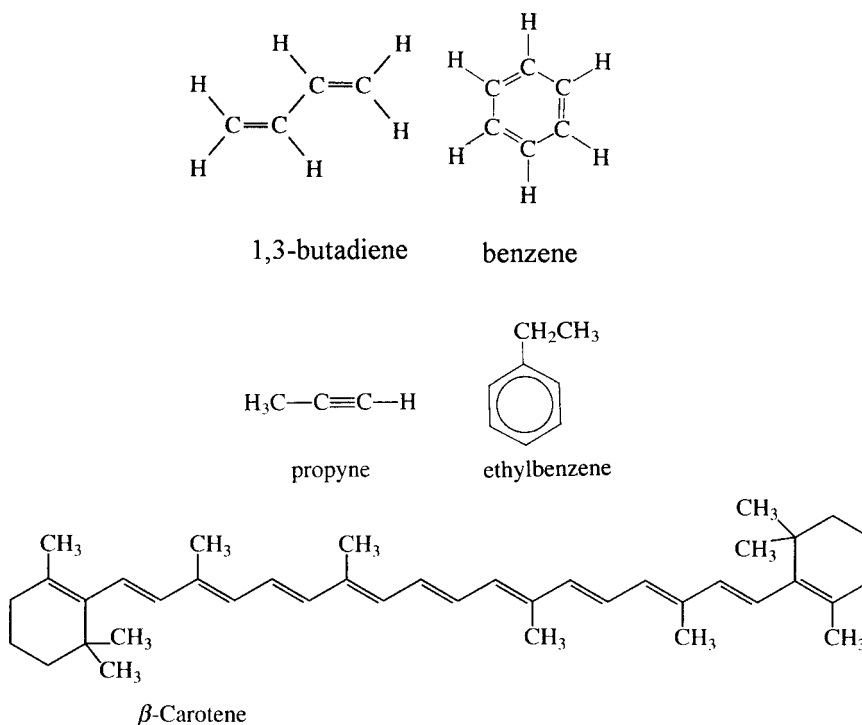
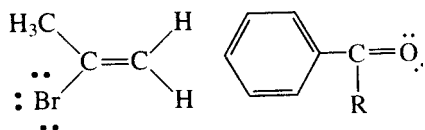
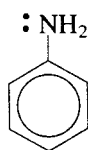


Figure 5.3 Examples of organic molecules containing π bonds. Note that benzene rings can be drawn showing three π bonds (the Kekulé structure) or with a circle inside the ring, as has been done for ethylbenzene, to more accurately depict the delocalized nature of the π electrons in aromatic compounds.



2-bromopropene



aminobenzene

Figure 5.4 Examples of organic molecules with nonbonding electrons. The n electrons are represented as pairs of dots around the atom on which they are located. For the carbonyl compound, if $R = H$, the compound is an aldehyde; if $R =$ an organic group, the compound is a ketone.

A schematic energy diagram of two s electrons in atomic orbitals on adjacent atoms combining to form a σ bond is shown in Fig. 5.5. Orbitals are conserved, therefore two molecular orbitals are formed, a sigma bonding orbital and a higher energy sigma antibonding orbital. The antibonding orbital is denoted σ^* . The energy difference between σ and σ^* is equal to ΔE , shown by the large arrow. Remember that each atom has three $2p$ atomic orbitals. One of those p orbitals can overlap with a p orbital on an adjacent atom to form a second set of sigma orbitals. Sideways overlap of the other two p orbitals is possible, resulting in pi bonding and antibonding orbitals. The schematic energy diagram for the formation of one set of π orbitals is shown in Fig. 5.6, and the energy difference between the π orbital and the antibonding π^* orbital is shown by the large arrow. If a p orbital is filled with a pair of electrons, it will have no tendency to form a bond. Figure 5.7 shows that a filled atomic p orbital (in the atom on the right) may form a nonbonding n orbital that is unshifted in energy from the atomic orbital, while

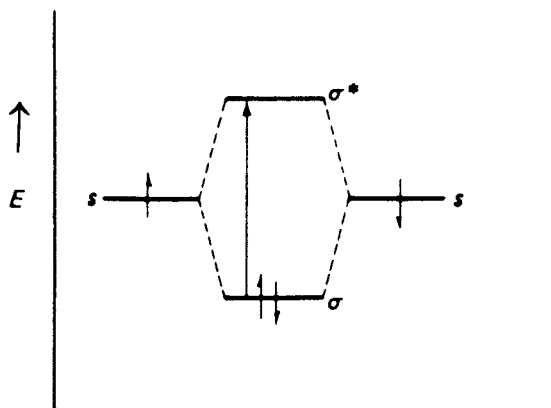


Figure 5.5 Schematic energy diagram of two s orbitals on adjacent atoms forming a σ bonding orbital and a σ^* antibonding orbital.

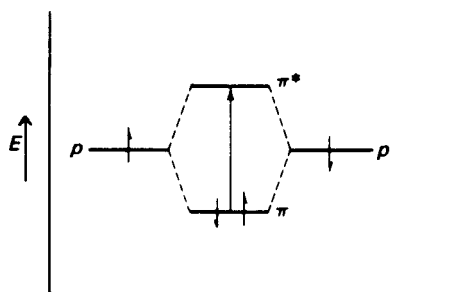


Figure 5.6 Schematic energy diagram of two p orbitals on adjacent atoms forming a π bonding orbital and a π^* antibonding orbital.

the partially filled p orbitals on each atom overlap to form a pair of pi bonding and anti-bonding orbitals.

A relative energy diagram of σ , π , and n electrons is shown in Fig. 5.8, although there are exceptions to this general order. It can be seen that the energy required to excite an electron from a σ to a σ^* orbital is considerably greater than that required to excite an electron from a π to a π^* orbital or an n electron to either a σ^* or a π^* orbital. As a consequence, the energy necessary to excite σ electrons to σ^* orbitals is greater than that available in the UV region, but usually UV radiation is sufficient to excite electrons in π orbitals to π^* antibonding orbitals or n electrons to π^* or σ^* antibonding orbitals.

5.1.2. Absorption by Molecules

Quantum mechanics provides a theoretical basis for understanding the relative energy levels of molecular orbitals and how they vary with structure. Quantum mechanics also generates a set of “selection rules” to predict what transitions occur in molecules. The transitions that occur in molecules are governed by quantum mechanical selection rules. Some transitions are “allowed” by the selection rules, while others are “forbidden”. The selection rules are beyond the scope of this text, but may be found in most physical chemistry texts or in the

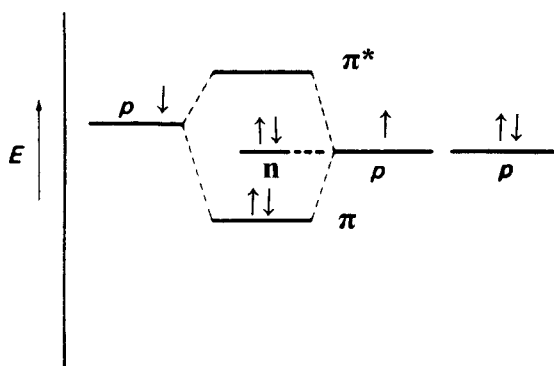


Figure 5.7 The relative energy levels of the π , π^* , and nonbonding (n) orbitals formed from p orbitals on adjacent atoms.

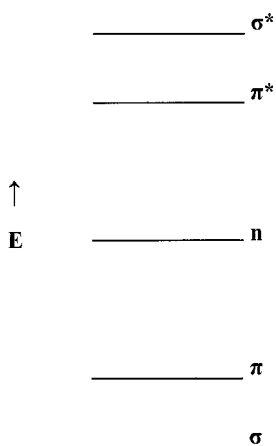


Figure 5.8 The relative energy levels of a set of σ , π , and n orbitals and the associated antibonding orbitals.

text by Ingle and Crouch listed in the bibliography. As is often the case with rules, there are exceptions, and many forbidden transitions do occur and can be seen in UV/VIS spectra.

When molecules are electronically excited, an electron moves from the highest occupied molecular orbital to the lowest unoccupied orbital, which is usually an antibonding orbital. Electrons in π bonds are excited to antibonding π^* orbitals, and n electrons are excited to either σ^* or π^* orbitals.

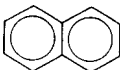
Both organic and inorganic molecules may exhibit absorption and emission of UV/VIS radiation. Molecular groups that absorb visible or UV light are called *chromophores*, from the Greek word *chroma*, color. For example, for a $\pi \rightarrow \pi^*$ transition to occur, a molecule must possess a chromophore with an unsaturated bond, such as $C=C$, $C=O$, $C=N$, and so on. Compounds with these types of chromophores include alkenes, amides, ketones, carboxylic acids, and oximes, among others. The other transition that commonly occurs in the UV/VIS region is the $n \rightarrow \pi^*$ transition, so organic molecules that contain atoms with nonbonded electrons should be able to absorb UV/VIS radiation. Such atoms include nitrogen, oxygen, sulfur, and the halogen atoms, especially Br and I. Table 5.2 presents some typical organic functional groups that serve as chromophores. Table 5.3 lists types of organic compounds and the wavelengths of their absorption maximum,

Table 5.2 Organic Functional Groups that can Absorb UV/VIS Radiation

Functional group	Chemical structure	Electronic transitions
Acetylenic	$-C\equiv C-$	$\pi \rightarrow \pi^*$
Amide	$-\text{CONH}_2$	$\pi \rightarrow \pi^*$, $n \rightarrow \pi^*$
Carbonyl	$>C=O$	$\pi \rightarrow \pi^*$, $n \rightarrow \pi^*$
Carboxylic acid	$-\text{COOH}$	$\pi \rightarrow \pi^*$, $n \rightarrow \pi^*$
Ester	$-\text{COOR}$	$\pi \rightarrow \pi^*$, $n \rightarrow \pi^*$
Nitro	$-\text{NO}_2$	$\pi \rightarrow \pi^*$, $n \rightarrow \pi^*$
Olefin	$>C=C<$	$\pi \rightarrow \pi^*$
Organoiodide	$R-I$	$n \rightarrow \sigma^*$
Thiol	$R-SH$	$n \rightarrow \sigma^*$

Note: R = any organic group (e.g., CH_3 , C_2H_5 , C_6H_5 , etc.).

Table 5.3 Absorption Wavelengths of Typical Organic Functional Groups

Chromophore	System	Wavelength of absorption maximum, λ_{max} (nm)
Amine	$-\text{NH}_2$	195
Bromide	$-\text{Br}$	208
Iodide	$-\text{I}$	260
Thioketone	$>\text{C}=\text{S}$	460
Thiol	$-\text{SH}$	220
Ester	$-\text{COOR}$	205
Aldehyde	$-\text{CHO}$	210
Carboxylic acid	$-\text{COOH}$	200–210
Nitro	$-\text{NO}_2$	210
Azo	$-\text{N}=\text{N}-$	285–400
Conjugated olefins	$(-\text{HC}=\text{CH}-)_2$	210–230
	$(-\text{HC}=\text{CH}-)_3$	260
	$(-\text{HC}=\text{CH}-)_5$	330
	$(-\text{HC}=\text{CH}-)_{10}$	460
Benzene		198
		255
Naphthalene		210
		220
		275

that is, the wavelength at which the most light is absorbed. Some compounds have more than one absorption peak, so several “maxima” are listed. Compounds such as alkanes (also called paraffins) contain only σ bonds, which do not absorb radiation in the visible or UV region.

Transition metal compounds are often colored, indicating that they absorb light in the visible portion of the spectrum. This is due to the presence of unfilled d orbitals. The exact wavelength of the absorption band maximum depends on the number of d electrons, the geometry of the compound, and the atoms coordinated to the transition metal.

5.1.3. Molar Absorptivity

Beer’s Law, which relates absorbance of a sample to the path length and concentration of absorbing species, was covered in Chapter 2. The proportionality constant, a , in Beer’s Law is the absorptivity of the absorbing species.

The *absorptivity*, a , of a molecule defines how much radiation will be absorbed by that molecule at a given concentration and at a given wavelength. If the concentration is expressed in molarity (mol/L, M), the absorptivity is defined as the *molar absorptivity*, ϵ . The absorptivity can be calculated directly from the measured absorbance using Beer’s Law:

$$A = abc = \epsilon bc \quad (5.2)$$

where A is the absorbance, b , the path length; and c , the concentration of the absorbing species.

If b is in units of cm and c has units of molarity, then the proportionality constant is the molar absorptivity and is given the symbol ϵ , with units of $\text{L mol}^{-1}\text{cm}^{-1}$. Commonly

$\epsilon \approx 10^4\text{--}10^5 \text{ L mol}^{-1} \text{ cm}^{-1}$ for an allowed transition and is on the order of 10–100 for a forbidden transition. The magnitude of the absorptivity is an indication of the probability of the electronic transition. High values of ϵ give rise to strong absorption of light at the specified wavelength; low values of ϵ result in weak absorption of light. Both a and ϵ are constants for a given wavelength and are physical properties of the molecule. The molar absorptivity may be specified for any wavelength, but is usually tabulated for the wavelength at which maximum absorption of light occurs for a molecule. The wavelength of maximum absorption is symbolized by λ_{max} and the associated ϵ is symbolized as ϵ_{max} . Table 5.4 presents typical values for λ_{max} and ϵ_{max} for some common organic molecules.

The absorptivity is not a direct measure of the probability that a given electronic transition will occur. This is because absorbance is measured over a wavelength range that is much smaller than the width of the absorption band. The absorptivity will differ at different wavelengths over the band profile. There are several fundamental quantities that are directly related to the transition probability; these include the transition probability, R^2 , Einstein coefficients, and the oscillator strength, f . The text by Ingle and Crouch presents the derivation of these quantities for the interested student.

5.1.4. The Shape of UV Absorption Curves

Figure 5.1 shows a “typical” UV absorption spectrum. The spectrum appears to be very simple, with a broad absorption “band” over a wide wavelength range instead of the numerous, narrower absorptions seen in IR spectra (Chapter 4). The absorption bands are broad because each electronic energy level has multiple vibrational and rotational energy levels associated with it. Excitation from the ground electronic state can occur to more than one vibrational level and to more than one rotational level. A schematic representation of an electronic transition with vibrational and rotational sublevels is shown in Fig. 5.9. In the ground state, only the lowest vibrational level is shown, with four rotational sublevels. At room temperature, most molecules are in the ground state in the lowest vibrational state. In the excited state, four vibrational sublevels are shown, slightly separated, with four rotational sublevels in each. Only four of the many possible transitions are shown; each arrow represents an absorption wavelength. The electronic transition consists of a large number of wavelengths that overlap to give the “continuous” absorption band observed. Even though each separate transition is quantized, the close energy spacing

Table 5.4 Typical Absorption Maxima and Molar Absorptivities for Common Chromophores

Chromophore	Compound	λ_{max} (nm)	ϵ_{max} ($\text{L mol}^{-1} \text{ cm}^{-1}$)
$>\text{C}=\text{C}-\text{C}=\text{C}<$	$\text{H}_2\text{C}=\text{C}-\text{C}=\text{CH}_2$	210	2.5×10^4
$-\text{NO}_2$	CH_3NO_2	210	1.0×10^4
		280	10
$-\text{N}=\text{N}-$	$\text{CH}_3\text{N}=\text{NCH}_3$	<250	$>1.0 \times 10^5$
$-\text{Br}$	CH_3Br	205	1.8×10^3
$-\text{SH}$	$\text{C}_2\text{H}_5\text{SH}$	230	160
Aromatic ring	Benzene (C_6H_6)	198	8.0×10^3
		255	200
	Naphthalene	221	1.0×10^5
		285	9.0×10^3
		300	290
$>\text{C}=\text{S}$	$(\text{CH}_3)_2\text{C}=\text{S}$	460	<10

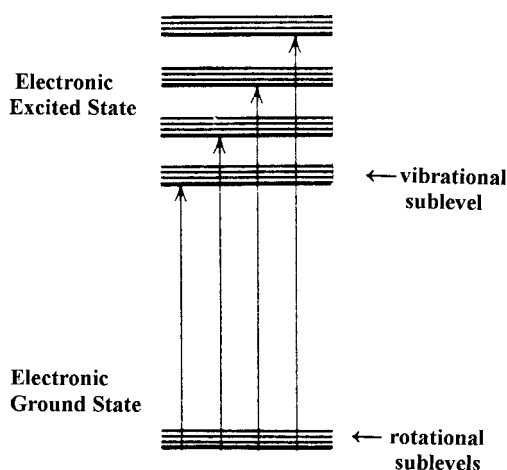


Figure 5.9 An electronic transition occurs over a band of energy due to the multiple vibrational and rotational sublevels associated with each electronic state. This schematic depicts four of the many possible transitions that occur. The length of the arrow is proportional to the energy required for the transition, so a molecular electronic transition consists of many closely spaced transitions, resulting in a band of energy absorbed rather than a discrete line absorption.

of the vibrational levels and the even more closely spaced rotational sublevels cause the electronic transition to appear as a broad band. This is shown schematically in Fig. 5.10.

An absorption band is characterized by its shape, that is, by its width and intensity. The shape of the band is determined primarily by the vibrational energy level spacing and the intensity of each vibrational transition. The intensity distribution is related to the probability of the transition to a given vibrational sublevel. The transition probabilities can be determined using the Franck–Condon principle. The texts by Hollas and Lambert et al. may be consulted for details. Suffice it to say that if we have a million molecules, even if they are mostly in the ground vibrational state before excitation, they may be in various vibrational states after excitation. The radiant energy required to cause electronic excitation to each vibrational energy level is slightly different and is further modified by rotational energy changes. For this reason, when UV radiation falls on the million molecules, it is absorbed at numerous wavelengths. The total range of the absorption wavelengths may stretch over 100 nm. The effect of the rotational energy of the molecule is to add even more

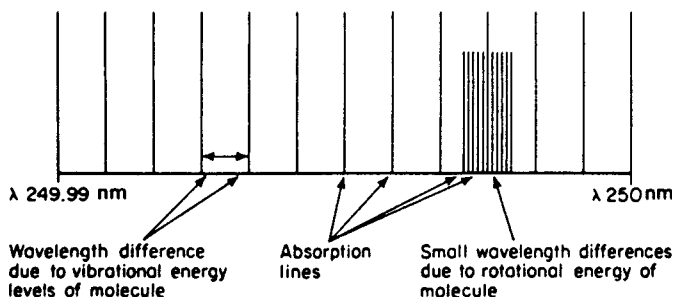


Figure 5.10 Illustration of a UV absorption band greatly expanded.

absorption lines to the single band. The increased number of absorption lines makes the lines even closer together, but it does not appreciably increase the total range of the band. This is because the energy involved in rotation is very small compared to vibrational energy and extremely small compared to electronic excitation energy. UV radiation is therefore absorbed in *absorption bands* rather than at discrete wavelengths.

In some cases, the UV/VIS spectra will show the different energies associated with the vibrational sublevels. For example, simple molecules in the gas phase often show the vibrational levels superimposed on the electronic transitions, as seen in Fig. 5.11, the gas phase spectrum for benzene. The sharp peaks on top of the broad bands are called vibrational “fine structure”. This fine structure is usually lost at high temperatures in the gas phase due to population of higher vibrational energy levels in the ground state electronic level, with the result that many more lines are seen. Molecules in solution (such as the spectrum shown in Fig. 5.1) usually do not exhibit vibrational structure due to interactions between the solvent and the solute molecules. Compare the gas phase spectrum of benzene (Fig. 5.11) to the solution spectrum for benzene (Fig. 5.12) and note the loss of much of the fine structure in solution. Fine structure due to rotational sublevels is never observed in routine UV/VIS spectra; the resolution of commercial instrumentation is not high enough to separate these lines.

5.1.5. Solvents for UV/VIS Spectroscopy

Many spectra are collected with the absorbing molecule dissolved in a solvent. The solute must be soluble in the solvent and the solvent must be transparent over the wavelength range of interest. A molecule will dissolve in a solvent if the formation of a solution leads to a lower energy system. The intermolecular attractive forces between the solute and solvent must be greater than solute–solute and solvent–solvent attractive forces. The forces involved in solution formation are dipole–dipole attraction, hydrogen bonding, and van der Waals forces. Polarity plays a major role, and gives rise to the “like dissolves like” rule. Polar substances dissolve more readily in polar solvents than

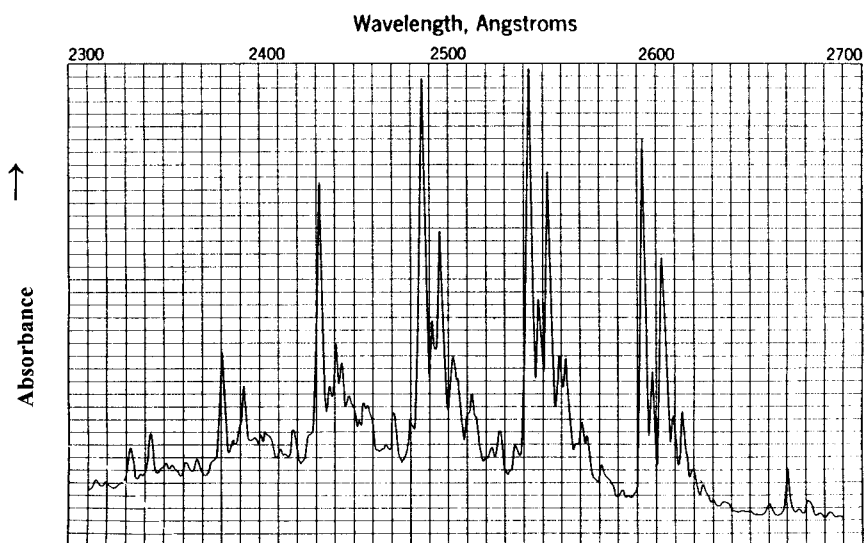


Figure 5.11 The gas phase absorption spectrum of benzene. (From Jaffé and Orchin. Reprinted with the permission of Professor M. Orchin.)

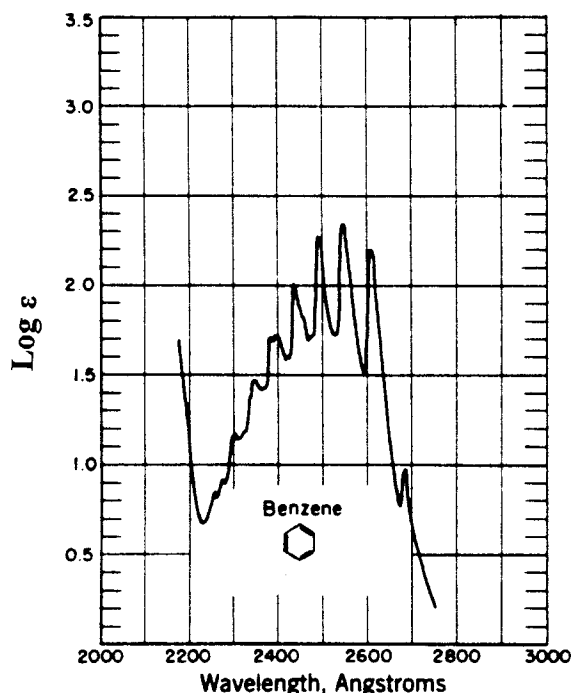


Figure 5.12 The absorption spectrum of benzene in solution. The solvent is cyclohexane. Note the loss of fine structure when compared to Fig. 5.11. (From Jaffé and M. Orchin. Reprinted with the permission of Professor M. Orchin.)

in nonpolar solvents. It is important for the solute to be dissolved completely; undissolved particles can scatter light from the light source. This can result in serious errors in qualitative and quantitative analyses.

The solvent may affect the appearance of the spectrum, sometimes dramatically. Polar solvents generally wipe out the vibrational fine structure in a spectrum. Solvents may also shift the position of the absorption band, as will be discussed. For the visible region of the spectrum any colorless solvent can be used in which the sample is soluble.

The common solvents used in UV/VIS spectroscopy are listed in Table 5.5, along with their low wavelength cutoff. At wavelengths shorter than the cutoff wavelength, the solvent absorbs too strongly to be used in a standard 1 cm sample cell. The cutoff is affected by the purity of the solvent. For spectroscopy, the solvents should be of spectral or spectrochemical grade, conforming to purity requirements set by the American Chemical Society.

5.2. INSTRUMENTATION

5.2.1. Optical System

Spectrometers are instruments that provide information about the intensity of light absorbed or transmitted as a function of wavelength. Both single-beam and the double-beam optical systems (see the schematics in Chapter 2) are used in molecular absorption spectroscopy. Single-beam systems and their disadvantages were discussed in Chapter 2. Most commercial instruments for absorption spectrometry are double-beam systems, so these will be reviewed.

Table 5.5 Common UV Solvents and their Lower Wavelength Cutoffs

Solvent	Lower wavelength cutoff (nm)
Acetone	330
Pyridine	306
Toluene	285
Carbon tetrachloride	265
Chloroform	245
Diethyl ether	215
Methanol	205
Cyclohexane	205
95% Ethanol	204
Hexane	195
Isooctane	195
Water	195
Acetonitrile	190

In the double-beam system, the source radiation is split into two beams of equal intensity. The two beams traverse two light paths identical in length; a *reference* cell is put in one path and the *sample* cell in the other. The intensities of the two beams after passing through the cells are then compared. Variation in radiation intensity due to power fluctuations, radiation lost to the optical system (e.g., cell surfaces, mirrors, etc.), radiation absorbed by the solvent, and so on should be equal for both beams, correcting for these sources of error. A dispersive spectrometer used for absorption spectroscopy that has one or more exit slits and photoelectric detectors that ratio the intensity of two light beams as a function of wavelength is called a spectrophotometer.

Commercial UV/VIS spectrometers are designed to operate with air in the light path over the range of 200–800 nm. Purging the spectrometer with dry nitrogen may permit wavelengths as low as 175 nm to be observed. For lower wavelengths, as mentioned, the spectrometer must be put under vacuum or purged with a nonabsorbing gas. Analytically, the vacuum UV region has been of minor importance for routine analysis because of the difficulties and expense inherent in instrumentation requiring a vacuum.

Simple optical systems using filters for wavelength selection and a photoelectric detector are called photometers. Photometers are used for both the visible and the UV region. For example, UV photometers were commonly used as detectors in HPLC but have been superseded by PDAs. HPLC detectors will be discussed in greater detail in Chapter 13.

All spectrometers for absorption measurements require a light source, a wavelength selection device, a sample holder, and a detector.

5.2.2. Radiation Sources

Radiation sources for molecular absorption measurements must produce light over a continuum of wavelengths. Ideally, the intensity of the source would be constant over all wavelengths emitted. Traditionally, the two most common radiation sources for UV/VIS spectroscopy were the tungsten lamp and the deuterium discharge lamp. The *tungsten lamp* is similar in functioning to an ordinary electric light bulb. It contains a tungsten filament heated electrically to white heat, and generates a continuum spectrum.

It has two shortcomings: the intensity of radiation at short wavelengths (<350 nm) is low; furthermore, to maintain a constant intensity, the electrical current to the lamp must be carefully controlled. However, the lamps are generally stable, robust, and easy to use. Typically, the emission intensity varies with wavelength as shown in Fig. 5.13. The shape of these curves is typical of the continuum output of a solid heated to incandescence. An incandescent solid that produces a curve of this type is called a blackbody radiator. The continuum emission is due to thermally excited transitions in the solid, in this case, the tungsten filament. The intensity vs. wavelength plot for a blackbody radiator is dependent on the temperature of the emitting material, not on its chemical composition. The tungsten lamp is most useful over the visible range and is therefore commonly used in spectrophotometry, discussed subsequently. Because it is used only in the visible region, the bulb (i.e., the lamp envelope) can be made of glass instead of quartz. Quartz is required for the transmission of UV light. The tungsten-halogen lamp, similar to the lamp in modern auto headlights, has replaced the older tungsten lamp in modern instruments. The tungsten-halogen lamp has a quartz bulb, primarily to withstand the high operating temperatures of the lamp. This lamp is much more efficient than a W lamp and has a significantly longer lifetime. The wavelength/intensity output of a tungsten halogen lamp is presented in Fig. 5.14.

The *deuterium arc lamp* consists of deuterium gas (D_2) in a quartz bulb through which there is an electrical discharge. The molecules are excited electrically and the excited deuterium molecule dissociates, emitting UV radiation. The dissociation of the deuterium molecule into atoms results in UV photon emission over a continuous range of energies from zero up to the energy of excitation of the molecule. This causes the lamp to emit a continuum (broadband) UV spectrum over the range of 160–400 nm rather than a narrow line atomic emission spectrum. The lamps are stable, robust, and widely used. The use of deuterium (D_2) instead of hydrogen gas results in an increase in the emission intensity

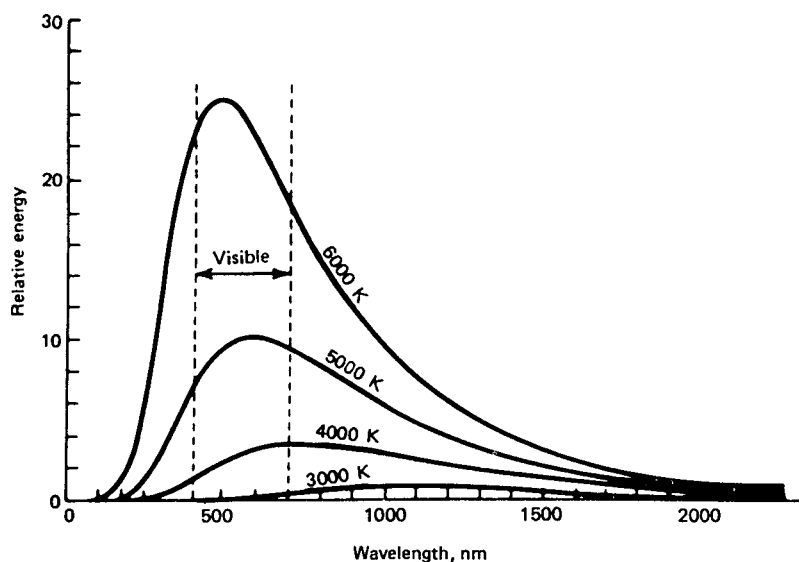


Figure 5.13 Emission intensity of blackbody radiation at various temperatures as a function of wavelength: 3000 K is equivalent to a tungsten filament lamp (an incandescent lamp); 6000 K is equivalent to a xenon arc lamp.

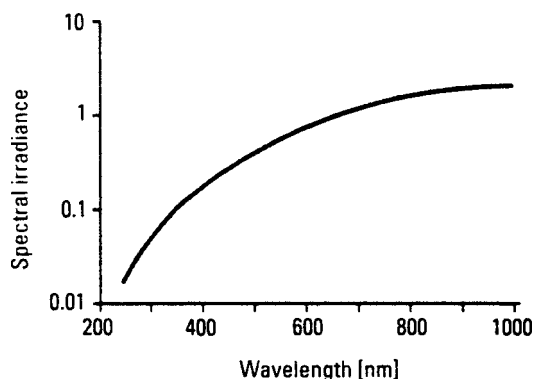


Figure 5.14 Emission spectrum of a commercial tungsten-halogen lamp. [Courtesy of Agilent Technologies (www.agilent.com).]

by as much as a factor of three at the short-wavelength end of the UV range. Deuterium is more expensive than hydrogen, but is used to achieve the high intensity required of the source. Figure 5.15 presents the emission spectrum of a deuterium arc lamp.

Xenon arc lamps operate in a manner similar to deuterium lamps. A passage of current through xenon gas produces intense radiation over the 200–1000 nm range. They provide very high radiation intensity and are widely used in the visible region and long-wavelength end of the UV range. This lamp is used in fluorescence spectrometry and the lamp schematic and spectrum are shown in Section 5.9.2.

5.2.3. Monochromators

The purpose of the monochromator is to disperse the radiation according to wavelength and allow selected wavelengths to illuminate the sample. Diffraction gratings are used to disperse light in modern instruments, as discussed in Chapter 2. The monochromators in modern systems, such as the Cary spectrophotometers from Varian, Inc. (www.varianinc.com) can scan at rates up to 2000–3000 nm/min, with slew rates (the time to move between wavelengths without taking measurements) as high as 16,000 nm/min to accommodate the high throughput measurements needed in pharmaceutical and biotechnology laboratories.

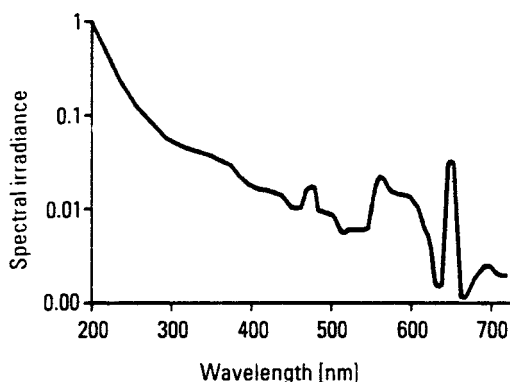


Figure 5.15 Emission spectrum of a commercial deuterium arc lamp. [Courtesy of Agilent Technologies (www.agilent.com).]

5.2.4. Detectors

The earliest detector used for visible light spectroscopy was the human eye. There are still *spectroscopes* and *color comparators* designed for visual observation of color and intensity.

Most modern instruments rely on **photoelectric transducers**, detection devices that convert photons into an electrical signal. Photoelectric transducers have a surface that can absorb radiant energy. The absorbed energy either causes the emission of electrons, resulting in a photocurrent or moves electrons into the conduction band of a solid semiconductor, resulting in an increase in conductivity. There are several common forms of these detectors including barrier layer cells, photomultiplier tubes, and semiconductor detectors.

5.2.4.1. Barrier Layer Cell

In a barrier layer cell, also called a photovoltaic cell, a current is generated at the interface of a metal and a semiconductor when radiation is absorbed. For example, silver is coated onto a semiconductor such as selenium (see Fig. 5.16) that is joined to a strong metal base, such as iron. To manufacture these cells, the selenium is placed in a container and the air pressure reduced to a vacuum. Silver is heated electrically, and its surface becomes so hot that it melts and vaporizes. The silver vapor coats the selenium surface, forming a very thin but evenly distributed layer of silver atoms. Any radiation falling on the surface generates electrons and holes at the selenium–silver interface. A barrier seems to exist between the selenium and the iron that prevents electrons from flowing into the iron; the electrons flow to the silver layer and the holes to the iron. The electrons are collected by the silver. These collected electrons migrate through an external circuit toward the holes. The photocurrent generated in this manner is proportional to the number of photons striking the cell.

Barrier layer cells are used as light meters in cameras and in low cost, portable instruments. The response range of these cells is 350–750 nm. These detectors have two main disadvantages: they are not sensitive at low light levels and they show *fatigue*, that is, the current drops gradually under constant exposure to light. On the plus side, they require no external electrical power and they are very rugged.

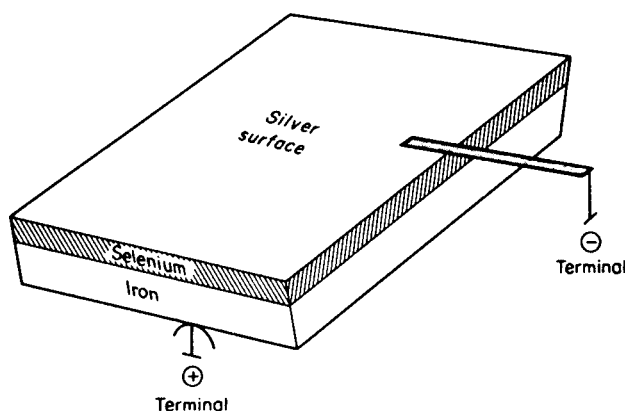


Figure 5.16 Barrier layer cell.

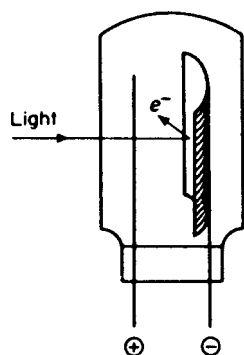


Figure 5.17 A vacuum phototube.

5.2.4.2. Photomultiplier Tube

The most common detector is the photomultiplier tube (PMT). A PMT is a sealed, evacuated transparent envelope (quartz or glass) containing a *photoemissive cathode*, an anode, and several additional electrodes called *dynodes*. The photoemissive cathode is a metal coated with an alkali metal or a mixture of elements (e.g., Na/K/Cs/Sb or Ga/As) that emits electrons when struck by photons. The PMT is a more sophisticated version of a vacuum phototube (Fig. 5.17), which contained only a photoemissive cathode and an anode; the photocurrent was limited to the electrons ejected from the cathode. In the PMT (Fig. 5.18), the additional dynodes “multiply” the available electrons. The ejected electrons are attracted to a dynode that is maintained at a positive

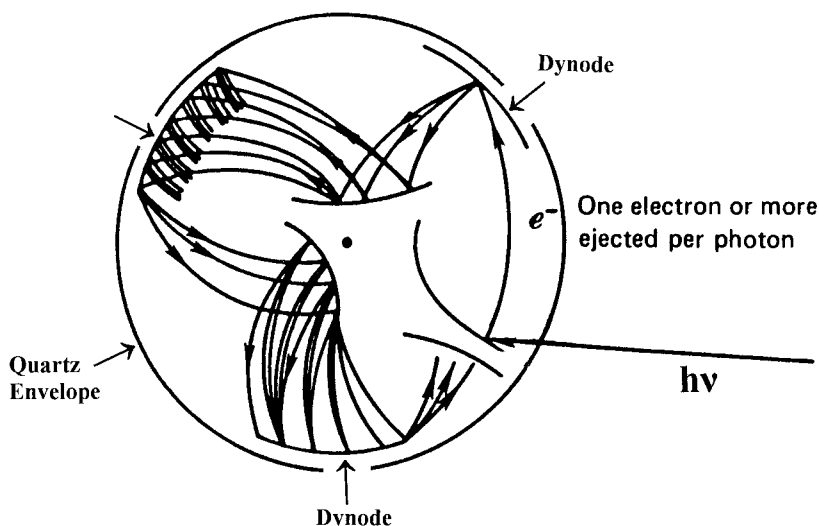


Figure 5.18 Schematic of a PMT, looking down through the tube. Impinging photons pass through the quartz envelope and liberate electrons from the light-sensitive cathode. The electrons are accelerated to the first dynode, where each electron liberates several electrons on impact. The process is repeated at the other dynodes, resulting in a cascade of electrons for every photon hitting the PMT.

voltage with respect to the cathode. Upon arrival at the dynode, each electron strikes the dynode's surface and causes several more electrons to be emitted from the surface. These emitted electrons are in turn attracted to a second dynode, where similar electron emission and more multiplication occurs. The process is repeated several times until a shower of electrons arrives at the anode, which is the collector. The number of electrons falling on the collector is a measure of the intensity of light falling on the detector. In the process, a single photon may generate many electrons and give a high signal. The dynodes are therefore operated at an optimum voltage that gives a steady signal. A commercial photomultiplier tube may have nine or more dynodes. The gain may be as high as 10^9 electrons per photon. The noise level of the detector system ultimately limits the gain. For example, increasing the voltage between dynodes increases the signal, but if the voltage is made too high, the signal from the detector becomes erratic or *noisy*. In practice, lower gains and lower noise levels may be preferable for accuracy.

PMTs are extremely sensitive to UV and visible radiation. In fact, they are so sensitive that care must be taken not to expose PMTs to bright light, to avoid damage. There are a wide variety of photoemissive surfaces available, which respond to different wavelength ranges. A plot of detector signal vs. wavelength is called a *response curve*. Figure 5.19 displays some response curves for commercial PMTs. The PMT detector should be chosen so that it has maximum response to the wavelength range of interest. For example, the IP28 is not useful at 800 nm, but the R136 and Ga/As PMT detectors respond in this range. PMTs have very fast response times, but they are limited in sensitivity by their *dark current*. Dark current is a small, constant signal from the detector when

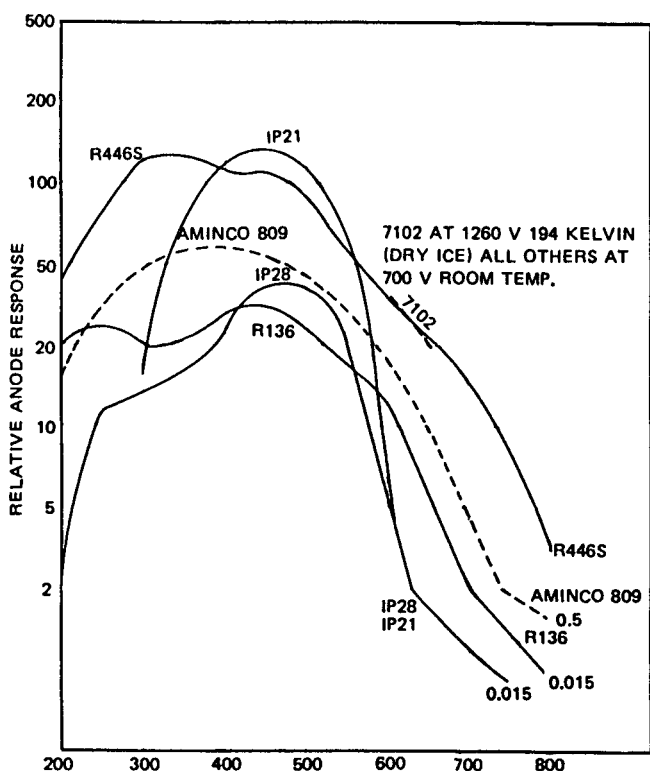


Figure 5.19 Response curves for various commercial PMTs. Note the variable response among different models and the sharp drop-off in response outside the useable range.

no radiation is falling on it. Dark current can be minimized or eliminated by cooling the detector housing. Cooling devices are available commercially for this purpose.

5.2.4.3. Semiconductor Detectors—Diodes and Diode Array Systems

Solid semiconducting materials are extremely important in electronics and instrumentation, including their use as radiation detectors. To understand the behavior of a semiconductor, it is necessary to briefly describe the bonding in these materials.

When a large number of atoms bond to form a solid, such as solid silicon, the discrete energy levels that existed in the individual atoms spread into *energy bands* in the solid. The valence electrons are no longer localized in space at a given atom. The width of the energy bands increases as the interatomic spacing in the solid decreases. The highest band that is at least partially occupied by electrons is called the *valence band*; the energy band immediately above the valence band is called the *conduction band*. The valence and conduction bands are separated by a forbidden energy range (forbidden by quantum mechanics); the magnitude of this separation is called the *band gap*, E_g . A set of energy bands and the band gap are shown schematically in Fig. 5.20. If the valence band of a solid is completely filled at a temperature of 0 K, the material is a **semiconductor** or an **insulator**. The difference between a semiconductor and an insulator is defined by the size of the band gap. If $E_g \leq 2.5$ eV, the material is a semiconductor; if $E_g > 2.5$ eV, the material is an insulator. The third type of material, a **conductor**, has a partially filled valence band at 0 K.

The two elements most used for semiconductor devices are silicon and germanium; both are covalently bonded in the solid state and both belong to group 4A of the periodic table. [This group is also called group 14 in the new International Union of Pure and Applied Chemistry (IUPAC) nomenclature and group IVA in some texts.] Other semiconductors include GaAs, CdTe, InP, and other inorganic and organic compounds. Most semiconductors are covalently bonded solids. Band gap energies for semiconductors are tabulated in the CRC Handbook of Chemistry and Physics.

Silicon has the valence electronic structure $3s^2 3p^2$. The partially filled p orbitals might lead one to suppose that silicon has a partially filled valence band and would therefore be an electrical conductor. Because silicon is covalently bonded, the two 3s electrons and the two 3p electrons occupy sp^3 hybrid orbitals. This results in a solid with two electron energy bands, each with four closely spaced sublevels, one for each electron in the valence shell of Si. The four electrons occupy and fill the valence band at 0 K and are therefore nonconducting. However, at temperatures above 0 K, a few electrons can

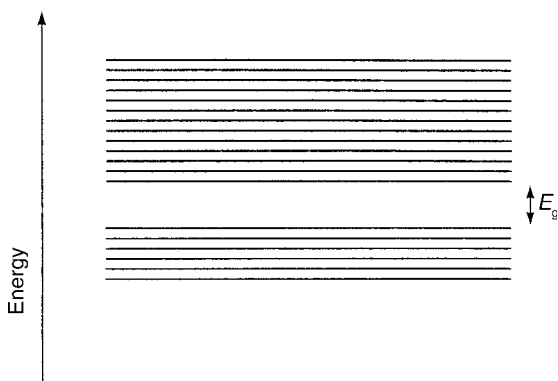


Figure 5.20 Schematic of energy bands in a solid material separated by a band gap of energy E_g .

be thermally promoted from the valence band into the conduction band; there they become conductors of electricity. When an electron leaves the valence band, it leaves behind a positive hole that is also mobile, thus producing an electron–hole pair. Both the electron and the hole are charge carriers in a semiconductor. Semiconductors such as Si and Ge are called intrinsic semiconductors; their behavior is a result of the band gap and band structure of the pure material.

The conductivity can be increased by doping either one of these elements with a group 5A element, such as arsenic or antimony, or a group 3A element, such as indium or gallium. Doping means to add another species to the *host* material; the added species is referred to as the *dopant*. The electrons associated with the dopant atom do not have the same energy levels as the host and may lie at energies forbidden to the host. Conductivity caused by addition of a dopant is called *extrinsic conduction*. A group 5A element has an extra electron (or extra negative charge). This electron is not held as tightly as the covalently bonded electrons of the host and requires less energy to move it into the conduction band. This is an n-type semiconductor. Similarly, adding a group 3A element leads to “missing” electrons, which can be considered to be the generation of extra positive holes. These positive holes from the dopant atom can accept electrons from the valence band. The energy needed to move an electron into an acceptor hole is less than the energy needed to move an electron into the conduction band. This is a p-type semiconductor. In an n-type semiconductor the electron is mobile, and in the p-type the positive hole is mobile. In an intrinsic semiconductor, two charge carriers are formed for every excitation event. In extrinsic semiconductors, either n or p type, only one charge carrier is formed per excitation event.

Semiconductors can be used as detectors for electromagnetic radiation. A photon of light with $E > E_g$ is sufficient to create additional charge carriers in a semiconductor. Additional charge carriers increase the conductivity of the semiconductor. By measuring the conductivity, the intensity of the light can be calculated. Selection of a material with the appropriate band gap can produce light detectors in the UV, visible, and IR regions of the spectrum.

5.2.4.4. Diodes

A diode or rectifier is an electronic device that permits current to flow in only one direction. If we put together a p-type semiconductor and an n-type semiconductor, the junction between the two types is a pn junction, as shown in Fig. 5.21. It is formed from a single piece of semiconductor by doping one side to be a p-type and the other side to be an n-type semiconductor. The junction is formed where the two types meet. Before any potential is applied to the device, holes will be the major charge carriers on the p side and electrons will be the major charge carriers on the n side. If we apply a positive potential to the p-type side and a negative potential to the n-type side, as shown in Fig. 5.22, positive charges (holes) flow from the p region to the junction and negative charges flow from the n region to the junction. At or near the junction, the holes and

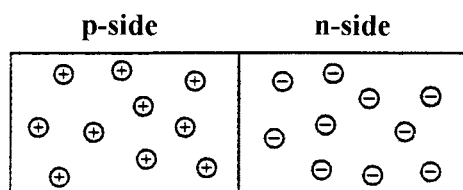


Figure 5.21 A p–n junction with no applied electrical potential.

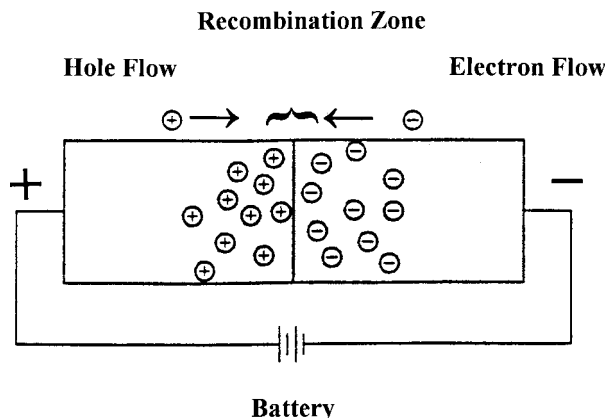


Figure 5.22 A p–n junction under forward bias. The positive battery terminal is connected to the p side.

electrons recombine and are annihilated. This is called a *forward bias*, and under these conditions current flows easily across the semiconductor. The annihilation of the electron–hole pair produces energy.

However, if the applied voltage were in the reverse direction, the flow of carriers would be in the opposite direction, as shown in Fig. 5.23(a). These are the conditions of *reverse bias*. The junction region is depleted of mobile charge carriers, recombination cannot occur, and no significant flow of current occurs. There is always a small flow of current due to the intrinsic conductivity. In short, the pn junction acts as a rectifier and permits significant current flow only under forward bias.

If a diode is held under reverse bias, and photons of energy greater than the band gap energy fall on the diode junction as shown in Fig. 5.23(b), electron–hole pairs are formed in the depleted region. These carriers will move through the diode, producing a current that is proportional to the intensity of the light falling on the diode. These detectors cover spectral ranges from the UV (about 190 nm) into the NIR (about 1000 nm), but are not as sensitive as PMTs. They have limited dynamic range compared to PMTs and when they deviate from linearity, they do so precipitously.

5.2.4.5. Diode Arrays

In UV/VIS spectroscopy a complete absorption spectrum can be obtained by scanning through the entire wavelength range and recording the spectrum with a PMT, one wavelength at a time. This takes time using a conventional scanning monochromator system, although modern instruments are much faster than old instruments. The absorption at one wavelength is measured at a different time from that at another wavelength.

There are two conditions under which scanning optical systems do not work very well. The first is when there is a rapid chemical reaction taking place and conventional scanning is too slow to follow the reaction. The second is when the sample is available only for a limited time and complete scanning is not possible. Examples of the latter include the eluent from a liquid chromatographic separation, the flowing stream in a flow injection system, or a process stream in a chemical or pharmaceutical production plant. In cases like this, many wavelengths need to be monitored simultaneously. Ideally, intensity over the entire spectral range of interest should be measured at the same instant.

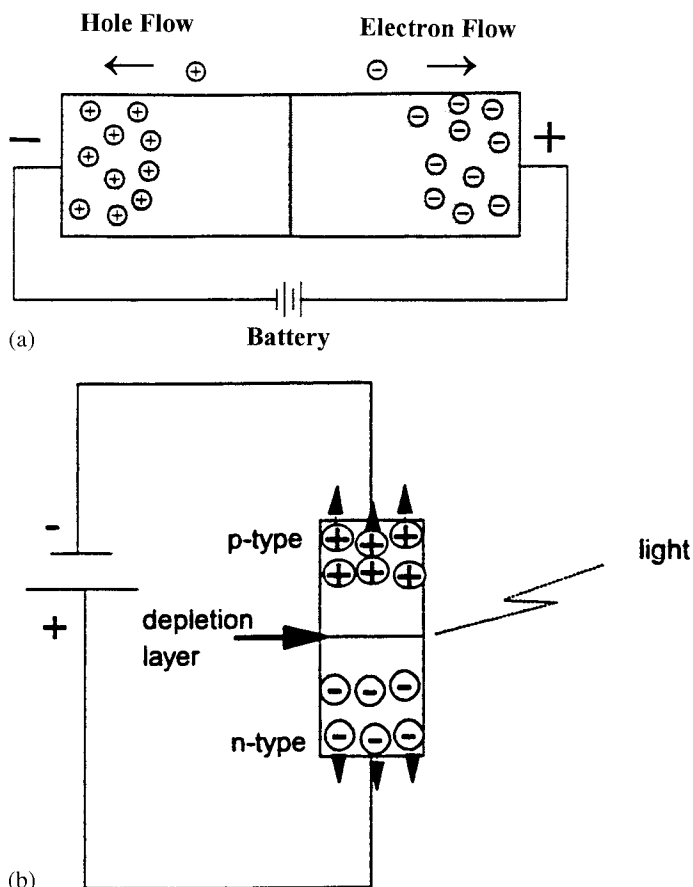


Figure 5.23 (a) A p–n junction under reverse bias. The positive battery terminal is connected to the n side. A depletion layer forms along the junction. (b) Diagram of a photodiode showing light incident upon the depletion layer. (From Brown, used with permission.)

The linear photodiode array (LPDA) is a transducer developed to enable simultaneous measurement of light intensity at many wavelengths. The diode array consists of a number of semiconductors embedded in a single crystal in a one-dimensional linear array. A common procedure is to use a single crystal of doped silicon that is an n-type semiconductor. A small excess of a group 3A element, such as arsenic, is embedded into the surface at regular intervals. This creates local p-type semiconductors. The semiconductor device ideally has a cross-section such as that shown in Fig. 5.24. The surface contains a linear series or array of pn junctions, each of which is a photodiode. The individual diodes are called *elements*, *channels*, or *pixels*.

The PDA is arranged as part of a circuit. A reverse bias is created across the pn junction by charging a capacitor in the circuit. Radiation falling on the array creates charge carriers in both p and n regions. The electrons will then flow to the nearest p-type semiconductor and the holes are collected in the p-type region. The current flow partially discharges the capacitor. At the end of the measurement cycle the capacitor is recharged; the charging current results in a voltage that is directly related to the light intensity. The number of charge carriers formed in a particular element depends on the light intensity falling on the array in the immediate vicinity of that particular element. By

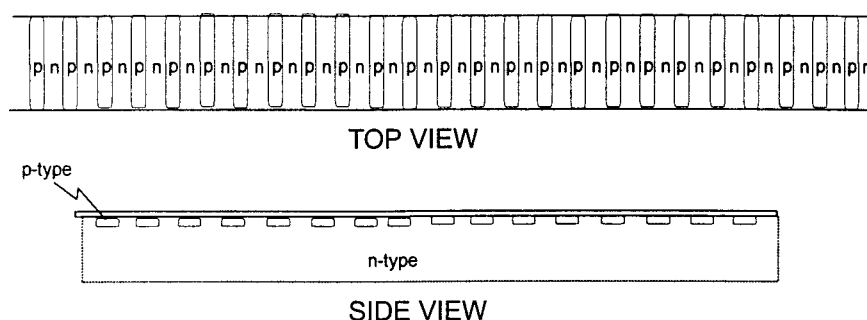


Figure 5.24 A PDA. The top view shows the face that the light would fall on. The side view shows that the p-type elements are embedded in a continuous layer of n-type semiconductor. (From Brown, used with permission.)

measuring the charges on each individual element, it is possible to get an instantaneous measurement of light intensity vs. wavelength of the whole spectral range, but in discrete elements. This is tantamount to a digital UV absorption spectrum.

The optical layout of a commercial multichannel instrument is shown schematically in Fig. 5.25. In this system radiation from a source, which may be a deuterium lamp or other UV/VIS light source, passes through the sample to a holographic grating, where the radiation is separated by wavelengths and directed to the diode array detector. No exit slit is used. The entire spectral region is measured in much less than 1 s. In practice, the spectrum is usually acquired for more than 1 s, and stored by the computer. This practice improves the signal-to-noise ratio for the measurement. By acquiring multiple measurements, the signal can be accumulated and sensitivity considerably increased.

PDA's are available to cover the wavelength range between 190 and 1100 nm. Simultaneous use of the entire wavelength range provides the multiplex advantage and improves the resolution of the system. The resolution of the system is limited by the

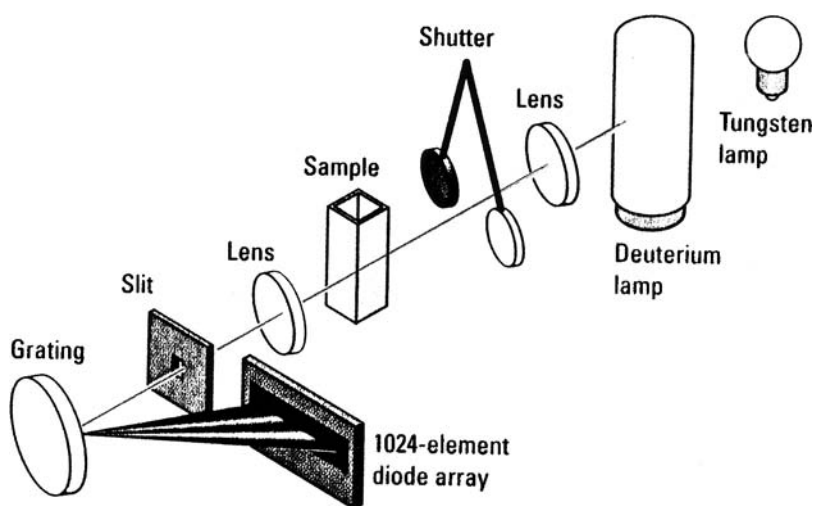


Figure 5.25 Optical diagram of the Agilent 8453 diode array spectrophotometer, showing both UV and visible light sources and a 1024 element PDA detector. [Courtesy of Agilent Technologies (www.agilent.com).]

number of diode elements involved. Typical diode spacing is 0.025 mm. Each one can be thought of as covering a finite spectral range. Detectors have been developed with as many as 4096 elements in the array, although 1024 is probably the most common number.

The most important applications for diode array systems are in molecular spectroscopy, since in general they do not have the resolution necessary for atomic spectroscopy. In molecular spectroscopy the most useful areas of application are for (1) scanning fast reactions to determine kinetics, (2) applications involving low light levels because spectra can be stored and added to each other, increasing the intensity, and (3) detectors for HPLC and capillary electrophoresis (CE). HPLC and CE are discussed in Chapter 13.

5.2.5. Sample Holders

Samples for UV/VIS spectroscopy can be solids, liquids, or gases. Different types of holders have been designed for these sample types.

5.2.5.1. Liquid and Gas Cells

The cells or *cuvettes* (also spelled *cuvets*) used in UV absorption or emission spectroscopy must be transparent to UV radiation. The most common materials used are quartz and fused silica. Quartz and fused silica are also chemically inert to most solvents, which make them sturdy and dependable in use. (*Note:* Solutions containing hydrofluoric acid or very strong bases, such as concentrated NaOH should never be used in these cells. Such solutions will etch the cell surfaces, making them useless for quantitative work.) Quartz and fused silica cells are also transparent in the visible and into the NIR region, so these could be used for all work in the UV and visible regions. These are also the most expensive cells, so if only the visible portion of the spectrum is to be used, there are cheaper cell materials available.

Some typical cell types are shown in Fig. 5.26. Cells are available in many sizes. The standard size for spectrophotometry is the 1 cm path length rectangular cell, which holds about 3.5 mL of solution, shown in the upper left of Fig. 5.26. There are microvolume cells (second from the left in the top row of Fig. 5.26) with volumes as small as 40 μL , flow-through cells for process streams or the routine analysis of large numbers of samples, microflow cells for chromatographic systems, and larger path length/volume cells for gases and highly dilute solutions. Two flow-through cells are shown in the middle of the bottom row in Fig. 5.26. In general, gas cells are long path cells, such as the one shown on the upper right of Fig. 5.26 and must be able to be closed once filled with a gas sample.

For spectrophotometric analysis in the visible region of the spectrum, glass or disposable plastic cells may be used. These are less expensive than quartz or fused silica but cannot be used at UV wavelengths. Plastic cells cannot be used with any organic solvent in which the plastic is soluble. Disposable plastic cells are not suitable for accurate quantitative work. Price differences are significant between the materials. For example, a high-quality 1 cm quartz cuvet for use in UV costs about \$70, the same size glass cuvet for use in the visible region costs about \$30 and a 1 cm plastic disposable cuvet costs about 10 cents. Microvolume cells, flow cells, and other specialty cells are expensive, with costs of \$200–500 per cell. Some spectrometers are designed to use ordinary glass test tubes as the “cells”. These test tube “cells” should not be used for accurate quantitative work. Transparencies of some typical cell materials are presented in Fig. 5.27.

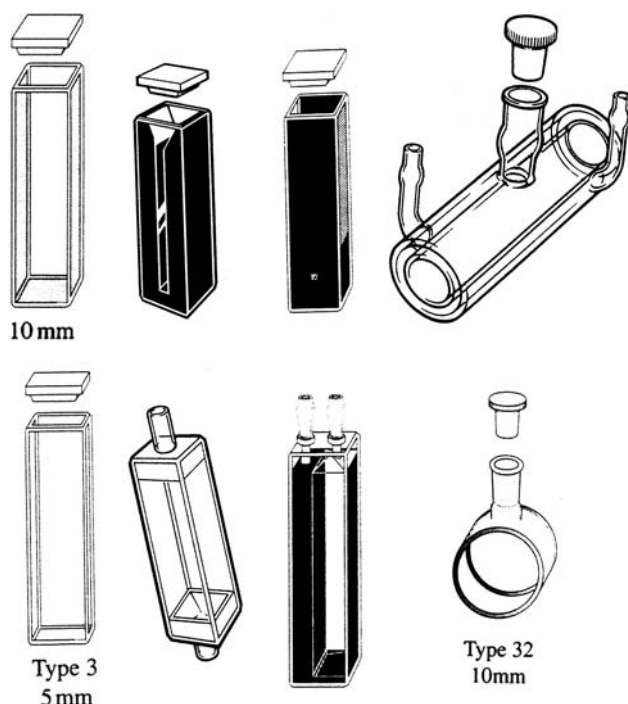


Figure 5.26 Cells for liquid samples, showing just a few of the wide variety of types and sizes available. From left to right, top row: standard 1 cm spectrophotometer cuvet with two optical faces and two frosted faces; semimicro 0.7 mL cuvet; 10 μ L submicro cell; constant temperature cell with a jacket for circulating a temperature-controlling fluid. From left to right, bottom row: 5 mm fluorometer cuvet (all four faces are optically clear); in-line continuous flow cell for process monitoring (sample flow is from bottom to top); 10 mm flow cell; cylindrical cell. [Courtesy of Starna Cells, Inc., Atascadero, CA (www.starna.com).]

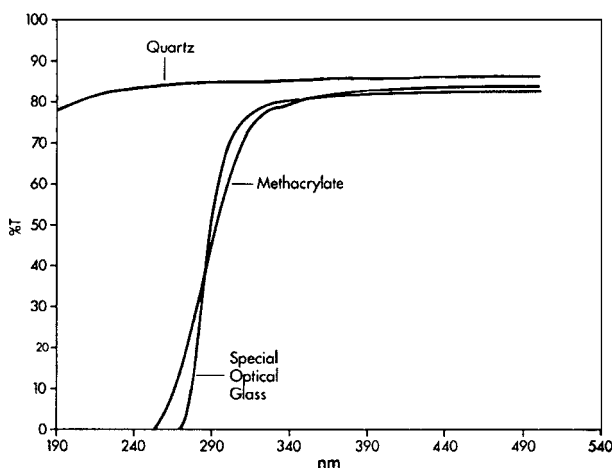


Figure 5.27 Transparencies of materials used to make sample cells for UV/VIS spectroscopy.

It is important that cells be treated correctly in order to achieve best results and to prolong their lifetime. To that end, the analyst should (1) always choose the correct cell for the analysis; (2) keep the cell clean, check for stains, etch marks, or scratches that change the transparency of the cell; (3) hold cells on the nontransparent surfaces if provided; (4) clean cells thoroughly before use and wash out the cell with a small amount of the sample solution before filling and taking a measurement; (5) not put strongly basic solutions or HF solutions into glass, quartz, or fused silica cells; (6) check for solvent compatibility with disposable plastic cells before putting them into the spectrometer; (7) for nondisposable cells, always dry carefully and return to their proper storage case; and (8) never wipe the optical surfaces with paper products, only lens cleaning paper or cloth material recommended by the manufacturer. At all times when not in use, cells should be kept clean and dry, and stored so that the optical surfaces will not become scratched.

5.2.5.2. *Matched Cells*

When double-beam instrumentation is used, two cells are needed: one for the reference and one for the sample. It is normal for absorption by these cells to differ slightly. This causes a small error in the measurement of the sample absorption and can lead to analytical error. For most accurate quantitative work, *optically matched cells* are used. These are cells in which the absorption of each one is equal to or very nearly equal to the absorption of the other. Large numbers of cells are manufactured at one time and their respective absorptivities measured. Those with very similar absorptivities are designated as optically matched cells. These are sold in pairs or sets of four. Matched cells are usually etched near the top with an identification mark and must be kept together. It is important for the analyst to understand that even closely matched cells will show small differences in absorption. The proper use of matched cells is to fill both the sample and the reference cells with the solvent and run a baseline spectrum, which is stored by the instrument computer system. The sample cell is then cleaned and sample solution put into it, while the reference cell and its solvent are left in place. After measuring the sample spectrum, the baseline is subtracted from the sample spectrum by the computer. This approach will correct for small differences in the cells. It is also important that the sample cell be reinserted facing in the same direction it was facing when the background was obtained. The etch mark on the top of the cell helps to facilitate this.

Modern cell manufacturing practices have improved greatly, and high-quality cell manufacturers are now producing cells with tolerances for window flatness, parallelity of windows, polish, and path length precision better than older “matched” cells. These modern cells are in effect optically matched by the nature of the manufacturing process. Tolerances for modern cells can be found on the Starna website (www.starnacells.com) as one example. It is still necessary for the analyst to check the cells on a routine basis by measuring a dilute solution of an absorbing material in all cells. This will identify any possible problems with microscopic scratches, residual film or deposits on the windows, and so on. Matched cells are not needed for qualitative analysis, such as obtaining the spectrum of a compound to help identify its structure.

5.2.5.3. *Flow-Through Samplers*

For routine analysis of large numbers of samples, the filling, cleaning, and emptying of large numbers of cells is time consuming. A flow cell and a peristaltic pump can be used to measure sample solutions directly from their original containers. This eliminates the need for sample handling and can minimize errors from sample handling, as well as eliminating

the need to wash many cuvetts. For example, Varian, Inc. makes a flow-through sampler for routine analysis for their Cary UV/VIS spectrometers that can send samples to a thermostatted flow cell as small as 80 μL sample volume. Dedicated flow-injection systems and segmented flow systems are available for specific routine analyses, such as nitrate, sulfate, and fluoride in drinking water. These systems are automated to take the sample, add and mix the reagents, and send the absorbing solution through a fixed wavelength spectrometer for completely unattended quantitative analysis.

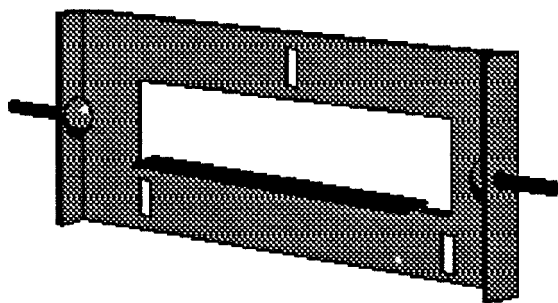
5.2.5.4. Solid Sample Holders

The absorption spectrum of thin films of transparent solid materials, such as polymers, can be obtained by using a film holder. The simplest type of holder can be a paper slide mount with the sample taped to the slide mount. However, producer of films, gels, and other sheet materials are often interested in the homogeneity of the film or sheet. The film holder accessory for the Cary line of spectrometers (Varian, Inc.) allows samples up to 160 mm in length to be mounted. The absorption spectrum can be collected and then the sample moved automatically to produce a plot of absorption vs. position along the sample length.

Gel electrophoresis is an important technique for separating high molecular weight biological molecules, such as DNA, lipoproteins, immunoglobulins, and enzyme complexes. The classic method for visualizing the separate molecules is to stain them with colored dyes. Slices of electrophoresis gels can be mounted in a holder (Fig. 5.28) and the absorption spectrum collected as a function of distance along the gel using the same device used to move film samples. The holder can be moved in increments of 0.25 mm and gels up to 100 mm long can be analyzed in this fashion.

5.2.5.5. Fiber Optic Probes

In all the cells described earlier, the sample had to be brought to the spectrometer and placed in the light path (or pumped into the light path). Modern fiber optics have enabled the spectrometer to be brought to the sample. Using a fiber optic probe such as the one shown in Fig. 5.29, the absorption spectrum can be collected from a very small sample volume in a microcentrifuge tube. Fiber optic probes can be used to collect a spectrum from inside almost any container—an open beverage can, a 55 gallon drum of material, a tanker truck, or railroad car full of liquid. Probes are made in various path lengths, just as cells are, but eliminate the need to collect a sample and put it into a cell for measurement. This is especially useful for unknown and possibly hazardous samples.



Gel Boat Holder

Figure 5.28 Gel boat holder for UV/VIS absorption spectroscopy of electrophoresis gels. [Courtesy of Varian, Inc., Walnut Creek, CA. (www.varianinc.com.)]

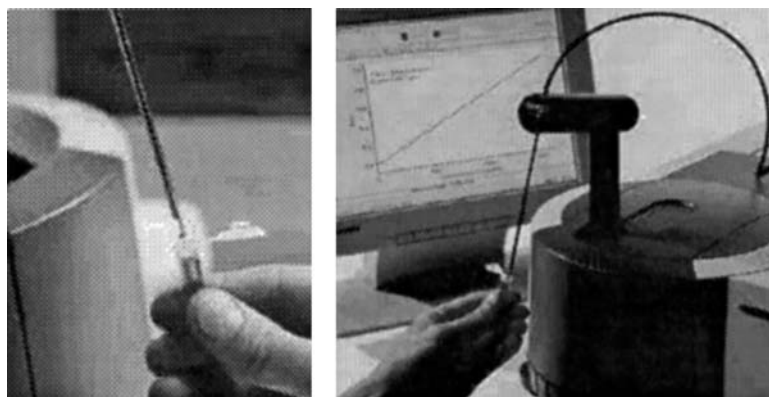


Figure 5.29 The photo on the left shows a fiber optics microprobe taking a spectrum from a 120 μL sample in a 500 μL tube. The photo on the right shows the fiber optics probe system for a Cary 50 spectrometer. [Courtesy of Varian, Inc., Walnut Creek, CA (www.varianinc.com).]

5.3. UV ABSORPTION SPECTRA OF MOLECULES

As has been discussed, the shape and intensity of UV/VIS absorption bands are related to the electronic structure of the absorbing species. This discussion will focus on the relationship of the absorption to the structure of simple organic molecules. Tables 5.3 and 5.4 list the approximate absorption maxima of common organic chromophores, functional groups that absorb UV and/or visible light. Of course, the chromophore is not an isolated unit; it is part of a molecule. The molecule is often dissolved in a solvent to acquire the spectrum. We will look at how we can use the absorption maximum of a chromophore and a set of guidelines to predict the position of the absorption maximum in a specific molecule. We will also consider how the solvent affects the spectrum of some molecules. As a reminder, the transitions that give rise to UV/VIS absorption by organic molecules are the $n \rightarrow \sigma^*$, $\pi \rightarrow \pi^*$, and $n \rightarrow \pi^*$ transitions.

Some terms need to be defined. A **chromophore** is a group of atoms (part of a molecule) that gives rise to an electronic absorption. An **auxochrome** is a substituent that contains unshared (nonbonding) electron pairs, such as OH, NH, and halogens. An auxochrome attached to a chromophore with π electrons shifts the absorption maximum to longer wavelengths. A shift to *longer* wavelengths is called a **bathochromic shift** or **red shift**. A shift to *shorter* wavelengths is called a **hypsochromic shift** or **blue shift**. An increase in the *intensity* of an absorption band (that is, an increase in ϵ_{max}) is called **hyperchromism**; a decrease in intensity is called **hypochromism**. These shifts in wavelength and intensity come about as a result of the structure of the entire molecule or as a result of interaction between the solute molecules and the solvent molecules.

5.3.1. Solvent Effects on UV Spectra

5.3.1.1. Bathochromic or Red Shift

There is a general observation that many molecules that absorb radiation due to a $\pi \rightarrow \pi^*$ transition exhibit a shift in the absorption maximum to a longer wavelength when the molecule is dissolved in a polar solvent compared to a nonpolar solvent. The shift to a longer wavelength is called a bathochromic or red shift. This does not mean that the solution turns red or that the absorption occurs in the red portion of the visible spectrum,

merely that the wavelength is shifted toward the red or longer wavelength end of the spectrum. This observation can be used to confirm the presence of $\pi \rightarrow \pi^*$ transitions in a molecule. The confirmation is carried out by dissolving a sample in two different solvents. For example, one solution can be made in a nonpolar solvent such as hexane and the second in a polar solvent such as alcohol. The spectra of the two solutions are recorded. If the absorption maximum in the alcohol solution occurs at a longer wavelength than the absorption maximum in the hexane solution, the compound exhibits a red shift. We would conclude that a $\pi \rightarrow \pi^*$ transition is present in the molecule and that the molecule must therefore have unsaturated bond(s). One major exception to this observation is the $\pi \rightarrow \pi^*$ transition in dienes and other hydrocarbon polyene molecules; the spectra of these molecules are not shifted by solvent polarity.

The reason for the shift in wavelength is related to the energy level of the excited state. The excited π^* state is more affected by attractive dipole–dipole interactions and hydrogen-bonding than the unexcited π state. Therefore, if a molecule is dissolved in a polar solvent, the energy level of the π^* antibonding orbital will decrease more than the energy level of the π bonding orbital. This is illustrated in Fig. 5.30. The energy difference in the polar solvent is less than the energy difference when the molecule is in a nonpolar solvent. As a consequence, the absorption maximum is changed to a longer wavelength in a polar solvent.

If the π^* energy level is decreased by attractive forces in polar solvents, it should be expected that the $n \rightarrow \pi^*$ transition will also show a red shift in polar solvents. It does, but there is a much more important interaction that overcomes the red shift for the $n \rightarrow \pi^*$ transition.

5.3.1.2. Hypsochromic or Blue Shift

The intermolecular attractive force known as hydrogen-bonding can occur when a molecule contains hydrogen covalently bound to oxygen or nitrogen. The O—H or N—H bond is highly polarized; the electrons in the covalent bond are pulled toward the electronegative atom, while the hydrogen can be thought of as having a partial positive charge. This hydrogen is able to form a “hydrogen bond” with an atom containing a pair of non-bonded electrons in an adjacent molecule. The hydrogen bond is the strongest of the intermolecular attractive forces. Examples of solvent molecules capable of hydrogen-bonding

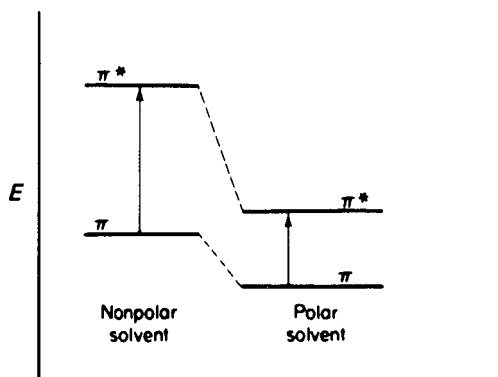


Figure 5.30 The energy difference between the π and π^* levels is decreased in the polar solvent. The absorption wavelength therefore increases. This is a bathochromic or red shift.

include water, alcohols such as ethanol and methanol, and molecules such as amines that contain an N—H bond.

The n electrons in a molecule are highly affected by hydrogen bond formation. The energy levels of n electrons decrease significantly in a solvent that has the ability to form hydrogen bonds. The result is an increase in the energy difference between the n orbital and the π^* orbital. This causes a shift in the absorption maximum of an $n \rightarrow \pi^*$ transition to shorter wavelengths, a blue shift, by as much as 25–50 nm. The decrease in the energy of the n electrons is almost equal to the energy of the hydrogen bond formed. The $n \rightarrow \pi^*$ transition also shows a hypsochromic shift as solvent polarity increases even in non-hydrogen-bonding solvents. This is thought to be due to the increased solvation of the n electrons; the energy level of the solvated electrons is lowered. The energy involved in the transition $n \rightarrow \pi^*$ when the solvent is nonpolar is less than the energy involved in the same transition when the molecule is in a polar or hydrogen-bonding solvent.

The hypsochromic shift effect on n electrons is much larger than the $n \rightarrow \pi^*$ bathochromic shift due to lowering of the π^* orbital described above. As a consequence, the absorption maximum for the $n \rightarrow \pi^*$ transition in a molecule which contains a lone pair of electrons will move to a shorter wavelength when dissolved in ethanol vs. hexane. Again, blue shift does not mean the absorption becomes blue, merely that the absorption wavelength becomes shorter. This information can be used to confirm the presence of n electrons in a molecule. The sample is dissolved in a non-hydrogen-bonding solvent such as hexane and also dissolved in a hydrogen-bonding solvent such as ethanol. If the absorption spectrum of the ethanol solution exhibits a blue shift (absorption maximum at a shorter wavelength) compared to the spectrum in hexane, n electrons are present in the sample molecule. Both $n \rightarrow \pi^*$ and $n \rightarrow \sigma^*$ transitions are blue-shifted by hydrogen bonding solvents. However, there are few $n \rightarrow \sigma^*$ transitions above 200 nm to begin with; a hypsochromic shift puts these absorptions further into the vacuum UV, so they are rarely observed in routine analysis.

A hypothetical example of how we can use this solvent effect information follows. A compound with molecules that contain both π and n electrons may exhibit two absorption maxima and may show both a red shift and a blue shift with a change in solvent polarity. In general, $\pi \rightarrow \pi^*$ transitions absorb approximately 10 times more strongly than $n \rightarrow \pi^*$ transitions. A molecule that contains both π and n electrons and is dissolved in a nonpolar solvent such as hexane might have an absorption spectrum similar to the spectrum in Fig. 5.31. By comparing the relative intensities of the two peaks, we would suspect that the absorption at 250 nm was due to a $\pi \rightarrow \pi^*$ transition and the absorption at 350 nm was due

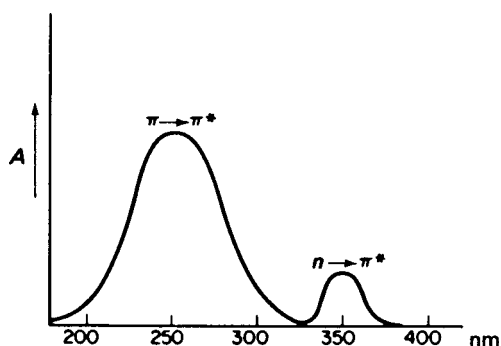


Figure 5.31 Expected absorption spectrum of a molecule in a nonpolar solvent exhibiting $\pi \rightarrow \pi^*$ and $n \rightarrow \pi^*$ transitions. Note the difference in absorptivity of the two transitions.

to an $n \rightarrow \pi^*$ transition. This is because the absorption coefficient of the $\pi \rightarrow \pi^*$ transition is considerably greater than that of the $n \rightarrow \pi^*$ transition, thus generating a higher degree of absorption. Also, in general, $n \rightarrow \pi^*$ transitions occur at longer wavelengths than $\pi \rightarrow \pi^*$ transitions because the energy difference for the $n \rightarrow \pi^*$ transition is lower.

If we now needed to confirm these assignments, we would put the compound in a solvent such as ethanol, which is both polar and capable of hydrogen-bonding. The polar nature of the solvent would induce a red shift in the $\pi \rightarrow \pi^*$ transition and hydrogen-bonding would induce a blue shift in the $n \rightarrow \pi^*$ transition. If our assignments were correct, then the absorption spectrum of the compound in ethanol might be as shown in Fig. 5.32. The combined evidence of the relative intensity (degree of absorption) and the blue and red shifts occurring in ethanol strongly supports the idea that the molecule contains both π bonds and n electrons.

5.4. UV SPECTRA AND THE STRUCTURE OF ORGANIC MOLECULES

Empirical rules based on thousands of laboratory observations have been developed over the years relating the wavelengths of the UV absorption maxima to the structures of molecules. R.B. Woodward in 1941 and then L. Fieser and M. Fieser developed rules for predicting the absorption maxima of dienes, polyenes, and conjugated ketones by studying terpenes, steroids, and other natural products. The rules are known as Woodward's Rules or the Woodward–Fieser Rules.

There are essentially four organic molecular systems of interest. The principal parent chromophore systems are (1) conjugated dienes, (2) monosubstituted benzene rings, (3) disubstituted benzene rings, and (4) conjugated carbonyl systems. The method of calculation is to identify a parent system and assign an absorption maximum. The parent system is then modified by the presence of other systems within the molecule. From these modifications, the absorption maximum of a specific molecular structure can be calculated.

5.4.1. Conjugated Diene Systems

The parent diene of conjugated diene systems is $C=C-C=C$. This system in a hexane solvent absorbs at 217 nm. If the conjugated system is increased, the wavelength of the

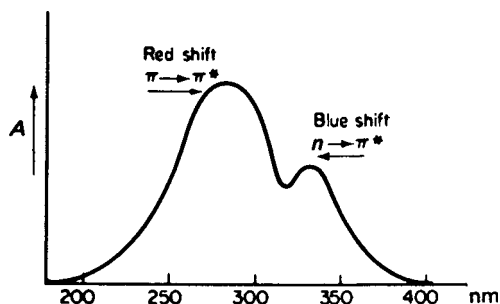


Figure 5.32 Expected absorption spectrum for the molecule in Fig. 5.31 dissolved in a polar solvent such as ethanol. There is a bathochromic (red) shift in the $\pi \rightarrow \pi^*$ transition and a hypsochromic (blue) shift in the $n \rightarrow \pi^*$ transition.

Table 5.6 Empirical Rules for Calculating the Absorption

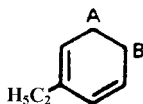
Maxima of conjugated dienes	nm
Absorption of parent diene system $C=C-C=C$	217
Shift to longer λ	
Double bond extension to diene system	30
Diene system within a ring	36
Exocyclic nature of double bond in conjugated system	5
Each alkyl substituent or ring residue	5
Auxochrome is	
O-acyl	0
O-alkyl	6
S-alkyl	30
N-alkyl ₂	60
Cl, Br	5

Source: With permission from A.I. Scott.

absorption maximum is increased by 30 nm for every double bond extension. Similarly, an alkyl group attached to the conjugated system increases the absorption maximum by 5 nm. Other substitutions, such as O-alkyl, cause an increase in the absorption maximum, as does inclusion within a ring or the exocyclic character of a double bond. These shifts are listed in Table 5.6. It should be emphasized that these results are empirical and not theoretical. They result from experimental observations.

Some applications of these rules are demonstrated. The first compound is shown in Example 5.1. We have a diene (the two double bonds) in a ring with an alkyl group attached to one of the diene carbon atoms. The predicted absorption maximum is calculated to be 268 nm. How was this done? Since we have a diene, we use Table 5.6. In Example 5.1, the value assigned to the parent diene is 217 nm. This diene is within a ring (a diene within a ring is called a “homoannular” diene); therefore we add 36 nm to the absorption maximum. It is not so clear that there are *three* alkyl substituents on the diene. One is obviously the ethyl group, $-C_2H_5$, but the “groups” in positions A and B are also alkyl substituents. In each case carbons at A and B are attached to the diene system and therefore contribute to the electron density and to the shift to a longer wavelength. The fact that carbons A and B are attached to each other does not change their effect on the shift of the absorption maximum. Each alkyl group adds 5 nm to the absorption maximum, so the peak should appear at $217 + 36 + 3(5) = 268$ nm.

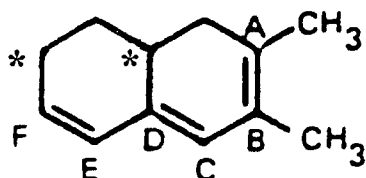
Example 5.1



Diene	217 nm
Within a ring	36
Alkyl substituent 3x5	15
Predicted $\lambda_{\max}$	268 nm

Example 5.2 is a little more complicated. The parent diene is at 217 nm. The diene system is within a ring, as in Example 5.1, so we add 36 nm. There is one double bond extension to the system. As you can see, the conjugated system consists of three double bonds rather than two double bonds. This adds

Example 5.2

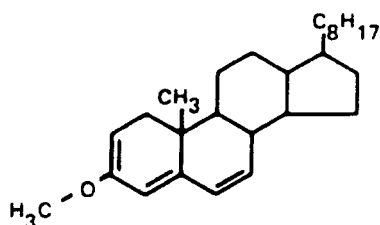


Diene	217 nm
Within a ring	36
Double bond extension	30
Exocyclic double bond	5
Alkyl substituents (5)	25
λ_{max}	313 nm

30 nm to the absorption maximum, from Table 5.6. There is one exocyclic double bond between carbons C and D, which adds 5 nm to the absorption maximum. This double bond is *touching* the adjacent ring and is within the conjugated system. The double bond between carbons E and F is also within a ring, but neither carbon *touches* the other ring, so this double bond is *not* exocyclic to a ring. The same thing is true for the double bond between carbons A and B; since neither carbon touches the adjacent ring, this bond is not exocyclic to a ring. There are five alkyl substituents as follows: two on carbon A, one on carbon B, one on carbon D, and one on carbon F. The alkyl substituents on carbons D and F are designated by asterisks. The five alkyl substituents add another 25 nm to the absorption maximum. So we predict the absorption maximum to be $(217 + 36 + 30 + 5 + 25) = 313$ nm.

Examples 5.3 and 5.4 are molecules with very similar formulas, but different structures. The double bonds are in different positions. The wavelength of the absorption maximum for Example 5.3 is 309 nm, which is 54 nm longer than in Example 5.4. The major difference between the two compounds is that the molecule in Example 5.3 has a two double bonds within one ring and a conjugated system of three double bonds, that is, the double bonds are each separated by one single bond. That results in the addition of 36 nm for the diene in a ring and 30 nm for the double bond extension. In Example 5.4, there are three double bonds, but the system is not conjugated. The double bond farthest to the right (on the upper part of the ring containing the two double bonds) is separated by two single bonds from the parent diene and is not part of a conjugated system.

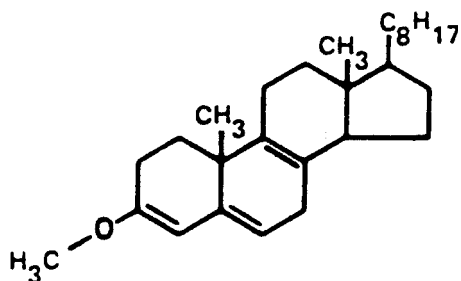
Example 5.3



Parent diene	217 nm
Within a ring (1)	36
Double bond extension (1)	30
Exocyclic double bond (1)	5
Alkyl substituents 3x5	15
$-\text{OCH}_3$	6
λ_{max}	309 nm

The only contributions to the absorption maximum in Example 5.4 are the exocyclic double bond (the diene bond on the right at the bottom of the ring containing the two double bonds) and the alkyl substituents. These two compounds can readily be distinguished by a simple UV absorption spectrum because of the large difference in λ_{max} .

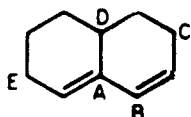
Example 5.4



Parent diene	217 nm
Within a ring (0)	—
Double bond extension (0)	—
Exocyclic double bond (1)	5
Alkyl substituents 3x5	15
$-\text{OCH}_3$	6
λ_{max}	243 nm

With the rules in Table 5.6, the absorption maximum in Example 5.5 would be calculated as follows:

Example 5.5

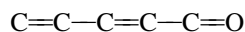


Parent diene	217 nm
One exocyclic double bond (A)	5
Three alkyl substituents (C, D, E)	15
Calculated λ_{max}	<u>237 nm</u>

The experimentally observed value is 235 nm. It should be noted that there is no homoannular diene system because the complete diene system is not contained in a single ring. Also, double bond B is not exocyclic to the ring, because it is not attached to a carbon that is part of another ring.

5.4.2. Conjugated Ketone Systems

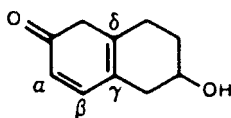
The parent system is



The absorption maximum assigned to this parent system is 215 nm. In a manner similar to that for conjugated dienes, the wavelengths of the absorption maxima for conjugated ketones are modified by extension of the double bond substitution and position relative to rings and relative to the carbonyl group. The carbons are labeled α , β , γ , and δ , and substitutions in these positions change the shift of the absorption maximum. The empirical values used for calculating the absorption maxima of different compounds are shown in Table 5.7.

An example of the calculation of the wavelength of the absorption maximum using Table 5.7 is shown in Example 5.6. In this example the calculations are similar to

Example 5.6

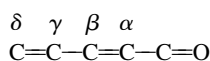


Parent ketone	215 nm
Alkyl substituent $\gamma(1)$	18
$\delta(2)$	36
Within a ring (1)	39
Double bond extension	<u>30</u>
Calculated λ_{max}	338 nm

those used in the conjugated diene system. Note that in order to affect the absorption maximum the substituents must be attached to the parent or conjugated system. The β carbon has two alkyl substituents and therefore increases the absorption maximum by 36 nm. The double bond between the γ and δ carbons is not an exocyclic double bond: it is within two different rings, but is not exocyclic to either of them. The OH group is *not* attached to the conjugated system and therefore does *not* contribute to its spectrum.

An isomer of the molecule used in Example 5.6 is shown in Example 5.7. The calculated absorption maximum is 286 nm. This spectrum is different in several ways from Example 5.6. For example, the carbon at the δ position has only one alkyl substitution.

Table 5.7 Rules for α,β -Unsaturated Ketone and Aldehyde Absorptions



Value assigned to parent α,β -unsaturated six-ring or acyclic ketone: 215

Value assigned to parent α,β -unsaturated five-ring ketone: 202

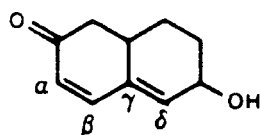
Value assigned to parent α,β -unsaturated aldehyde: 207

Increments added for	Shift to longer λ (nm)
Each exocyclic double bond	5
Diene within a ring	39
Double bond extending the conjugation	30
Each alkyl substituent	
α	10
β	12
γ and higher	18
Each	
OH	
α	35
β	30
δ	50
O—Acyl	
α, β, δ	6
O—Me	
α	35
β	30
γ	17
δ	31
S-alkyl	
β	85
Cl	
α	15
β	12
Br	
α	25
β	30
NR ₂	
β	95

Source: With permission from A.I. Scott.

The double bond between the γ and δ positions is exocyclic to a ring in this case and therefore increases the wavelength of the absorption maximum. The conjugated system is not within a ring, and therefore contributions from ring currents are not observed. These two isomers could be distinguished from each other by their UV spectra.

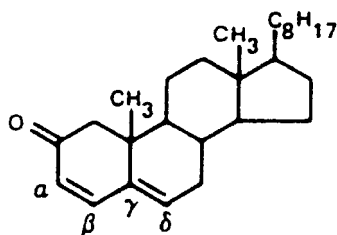
Example 5.7



Parent ketone	215 nm
Alkyl substituents $\gamma(1)$	18
$\delta(1)$	18
Within a ring	0
Double bond extension	30
Exocyclic to a ring	5
Calculated $\lambda_{\max}$	286 nm

The calculation of the absorption wavelength maximum for Example 5.8 is shown to be 286 nm. The calculation follows the steps used in previous examples. It should again be emphasized that the parts of the molecule that do not touch the conjugated or parent system do not shift the absorption maximum. This is true even for complex molecules, as some of these examples demonstrate. This makes prediction of the absorption maximum easier but the absorption maximum gives no information as to the structure of the entire molecule.

Example 5.8



Parent ketone	215 nm
Alkyl substituents $\gamma(1)$	18
$\delta(1)$	18
Exocyclic to a ring	5
Double bond extension	30
Calculated $\lambda_{\max}$	286 nm

The idea behind being able to predict absorption maxima based on structure is the hope that the absorption spectrum will tell us about the structure of an unknown. Examples 5.7 and 5.8 demonstrate the shortcoming of using the UV absorption maximum to deduce molecular structure. These are two very different compounds, easily distinguished by MS or NMR, but each gives the same UV absorption maximum. In fact, one could draw many different organic molecules that would give the same predicted absorption maximum. The UV absorption spectrum is only one of many tools needed to elucidate the structure of an organic molecule.

5.4.3. Substitution of Benzene Rings

Benzene is a strong absorber of UV radiation and particularly in the gas phase shows considerable fine structure in its spectrum (Fig. 5.11). Substitution on the benzene ring causes a shift in the absorption wavelengths. It is not uncommon for at least two bands to be observed, and frequently several more. The observed wavelengths of the absorption maxima of some substituted benzene rings are given in Table 5.8. These are experimental data and may be insufficient to completely identify unknown compounds. If the benzene ring is disubstituted, then calculations are necessary to predict the absorption maximum,

Table 5.8 Absorption Maxima of Monosubstituted Benzene Rings Ph—R

R	λ_{max} (nm) (solvent H ₂ O or MeOH)	
	Band 1	Band 2
—H	203	254
—NH ₃ ⁺	203	254
—Me	206	261
—I	207	257
—Cl	209	263
—Br	210	261
—OH	210	270
—OMe	217	269
—SO ₂ NH ₂	217	264
—CO ₂	224	268
—CO ₂ H	230	273
—NH ₂	230	280
—O [−]	235	287
—NHAc	238	
—COMe	245	
—CH=CH ₂	248	
—CHO	249	
—Ph	251	
—OPh	255	
—NO ₂	268	
—CH'=CHCO ₂ H	273	
—CH'=CHPh	295	

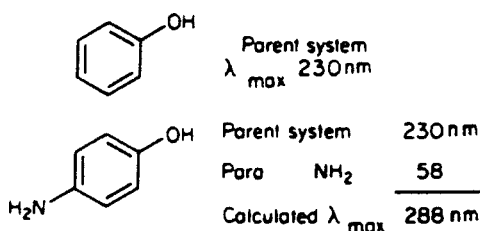
Note: Me = methyl, Ph = phenyl.

Source: From H.H. Jaffé and M. Orchin. Used with permission of Professor Milton Orchin.

because a list containing all the possible combinations would be very long and unwieldy and would need further experimental supporting evidence. There are some rules that help us understand disubstitution of benzene rings. These are as follows.

1. An electron-accepting group, such as NO_2 , and an electron-donating group, such as OH , situated ortho or para to each other tend to cancel each other out and provide a spectrum not very different from the monosubstituted benzene ring spectrum (Table 5.9).
2. Two electron-accepting groups or two electron-donating groups para to each other produce a spectrum little different from the spectrum of the monosubstituted compound.
3. An electron-accepting group and an electron-donating group para to each other cause a shift to longer wavelengths. A para-disubstituted benzene ring illustrating this is shown in Example 5.9.

Example 5.9



To become an expert in the interpretation of UV spectra requires reading of more detailed texts, as well as practice. It can be seen that the absorption maxima can be calculated with reasonable accuracy if the structure is known, usually within 5 nm or so. A complete study of this subject is beyond the scope of this book. More detailed treatments can be found in the texts by Creswell and Runquist, Pavia, Lampman and Kriz, or Silverstein, Bassler and Morrill (5th edition or earlier) listed in the bibliography.

5.5. ANALYTICAL APPLICATIONS

5.5.1. Qualitative Structural Analysis

As described at the beginning of the chapter, the types of compounds that absorb UV radiation are those with nonbonded electrons (n electrons) and conjugated double bond systems (π electrons) such as aromatic compounds and conjugated olefins. Unfortunately, such compounds absorb over similar wavelength ranges, and the absorption spectra overlap considerably. As a first step in qualitative analysis, it is necessary to purify the sample to eliminate absorption bands due to impurities. Even when pure, however, the spectra are often broad and frequently without fine structure. For these reasons, UV absorption is much less useful for the qualitative identification of functional groups or particular molecules than analytical methods such as MS, IR, and NMR. UV absorption is rarely used for organic structural elucidation today in modern laboratories because of the ease of use and power of NMR (Chapter 3), IR (Chapter 4) and MS (Chapters 9 and 10).

When UV spectra are used for qualitative identification of a compound, the identification is carried out by comparing the unknown compound's absorption spectrum

Table 5.9 Absorption Maxima of Disubstituted Benzene Derivatives

Substituent	Orientation	Shift for each substituent, γ , in EtOH (nm)
X = alkyl or ring residue		246
X = H		250
X = OH or O-alkyl		230
R = alkyl or ring residue	o-, m-	3
	p-	10
R = OH, O—Me, O-alkyl	o-, m-	7
	p-	25
R = O	o-	11
	m-	20
	p-	78
R = Cl	o-, m-	0
	p-	10
R = Br	o-, m-	2
	p-	15
R = NH ₂	o-, m-	13
	p-	58
R = NH—Ac	o-, m-	20
	p-	45
R = NH—Me	p-	73
R = NMe ₂	o-, m-	20
	p-	85

Note: o = ortho, m = meta, p = para; Me = methyl.

Source: From H.H. Jaffé and M. Orchin. Used with permission of Professor Milton Orchin.

with the spectra of known compounds. Compilations of UV absorption spectra in electronic and printed formats can be found from commercial sources such as the Informatics Division, Bio-Rad Laboratories (www.bio-rad.com) or the American Petroleum Institute (API) indices. Computer searching and pattern matching are the ways spectra are compared and unknowns identified in modern laboratories. Many academic libraries still maintain the printed spectra collections, which must be searched manually.

One area where qualitative UV/VIS spectroscopy is still very useful is in the rapid screening of samples for UV absorbing compounds. In a high throughput environmental lab, UV absorption can be used qualitatively to screen for samples that may have high levels of organic compounds, to avoid contaminating a sensitive instrument, or to evaluate dilution factors needed for quantitative analysis. In today's very high throughput pharmaceutical laboratories, UV absorption spectra can provide a quick survey of synthesized compounds, to screen out those that do not have the expected absorbance and therefore probably do not have the desired structure.

5.5.2. Quantitative Analysis

UV and visible absorption spectrometry is a powerful tool for quantitative analysis. It is used in chemical research, biochemistry, chemical analysis, and industrial processing. Quantitative analysis is based on the relationship between the degree of absorption and the concentration of the absorbing material. Mathematically, it is described for many

chemical systems by Beer's Law, $A = abc$, as discussed in Chapter 2. The term applied to quantitative absorption spectrometry by measuring intensity ratios is **spectrophotometry**. The use of spectrophotometry in the visible region of the spectrum used to be referred to as *colorimetry*. (The term colorimetry appears in much of the older literature, but the term is also used for an unrelated measurement, the specification of color in samples using the tristimulus system.) To avoid confusion, the term spectrophotometry should be used for both UV and visible regions when quantitative determination of an analyte species is meant. While qualitative analysis is most useful for organic molecules and some transition and rare earth compounds, quantitative UV/VIS spectrophotometry is useful for determination of organic molecules, inorganic molecules, metal and nonmetal ions, and organometallic complexes.

UV/VIS spectrophotometry is a widely used spectroscopic technique. It has found use everywhere in the world for research, clinical analysis, industrial analysis, environmental analysis, and many other applications. Some typical applications of UV absorption spectroscopy include the determination of (1) the concentrations of phenol, nonionic surfactants, sulfate, sulfide, phosphates, fluoride, nitrate, a variety of metal ions, and other chemicals in drinking water in environmental testing; (2) natural products, such as steroids or chlorophyll; (3) dyestuff materials; and (4) vitamins, proteins, DNA, and enzymes in biochemistry.

Quantitative UV/VIS spectrophotometry has been used for the determination of impurities in organic samples, such as in industrial plant streams using flow-through cells. For example, it can be used to determine traces of conjugated olefins in simple olefins or aromatic impurities in pure hexane or similar paraffins. It has also been used in the detection of possible carcinogenic materials in foods, beverages, cigarette smoke, and air. In the field of agriculture, UV/VIS spectrophotometry is used for the determination of nitrogen- and phosphorus-containing fertilizers. In the medical field, it is used for the determination of enzymes, vitamins, hormones, steroids, alkaloids, and barbiturates. These measurements are used in the diagnosis of diabetes, kidney damage, and myocardial infarction, among other ailments. In the pharmaceutical industry, it can be used to measure the purity of drugs during manufacture and the purity of the final product. For example, aspirin, ibuprofen, and caffeine, common ingredients in pain relief tablets, all absorb in the UV and can be determined easily by spectrophotometry.

Spectrophotometry is used routinely to determine the concentrations of metal and non-metal ions in a wide variety of samples. Spectrophotometry in the UV region of the spectrum is used for the direct measurement of many organic compounds, especially those with aromatic rings and conjugated multiple bonds. There are also colorless inorganic species that absorb in the UV. A good example is the nitrate ion, NO_3^- . A rapid screening method for nitrate in drinking water is performed by measuring the absorbance of the water at 220 and at 275 nm. Nitrate ion absorbs at 220 but not at 275 nm; the measurement at 275 nm is to check for interfering organic compounds that may be present. Spectrophotometric analysis in the visible region can be used whenever the sample is colored. Many materials are inherently colored without chemical reaction (e.g., inorganic ions such as dichromate, permanganate, cupric ion, and ferric ion) and need no further chemical reaction to form colored compounds. Colored organic compounds, such as dyestuffs, are also naturally colored. Solutions of such materials can be analyzed directly. The majority of metal and nonmetal ions, however, are colorless. The presence of these ions in a sample solution can be determined by first reacting the ion with an organic reagent to form a strongly absorbing species. If the product of the reaction is colored, absorbance can be measured in the visible region; alternatively, the product formed may be colorless but absorb in the UV. The majority of spectrophotometric determinations result in an increase in absorbance

(darker color if visible) as the concentration of the analyte increases. However, there are analyses that cause a bleaching of color (decrease in absorbance) with increasing concentration of analyte.

As was mentioned in the introduction, the color we observe in a colored solution is the color that is transmitted by the solution. The color absorbed is the complementary color. The relationship between the color of light absorbed and the color observed is given in Table 5.10.

There are thousands of possible compounds and complexes that can be formed by reacting analyte species with organic reagents. Ideally, the reagent chosen should be selective; that is, it should react with only one ion or molecule under the conditions present. Second, the reagent should cause an abrupt color change or absorbance change when mixed with the analyte. This imparts high sensitivity to the method. Third, this intensity of color or UV absorbance should be related to the concentration of ions in the sample. Spectrophotometric reagents have been developed for almost all metal and nonmetal ions and for many molecules or classes of molecule (i.e., for functional groups). Many of these reactions are both sensitive and selective. Several examples of these reagents and their uses are given in Table 5.11. The books by Boltz and by Sandell and Onishi listed in the bibliography are classic reference sources, but may be hard to locate. The analytical literature contains thousands of direct and indirect methods for quantitative analysis of metals and nonmetals. A good summary of methods with literature references for most metal and nonmetal ions may be found in the handbook by Dean listed in the bibliography.

Quantitative analysis by absorption spectrophotometry requires that the samples be free from particulates, that is, free from **turbidity**. The reason for this is that particles can scatter light. If light is scattered by the sample away from the detector, it is interpreted as an absorbance. The absorbance will be erroneously high if the sample is turbid. We can make use of the scattering of light to characterize samples as discussed in Section 5.7, but particulates must be avoided for accurate absorbance measurements.

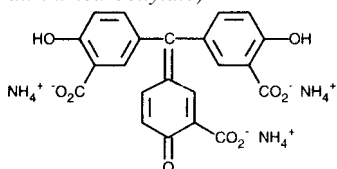
Quantitative analysis by spectrophotometry generally requires the preparation of a calibration curve, using the same conditions of pH, reagents added, and so on for all of the standards, samples, and blanks. It is critical to have a reagent blank that contains everything that has been added to the samples (except the analyte). The absorbance is measured for all blanks, standards, and samples. The absorbance of the blank is subtracted from all other absorbances and a calibration curve is constructed from the standards. The concentrations of analyte in the samples are determined from the calibration curve. The highest accuracy results from working in the linear region of the calibration curve. These quantitative methods can be quite complicated in the chemistry involved, the number of steps required (extraction, back-extraction, pH-adjustment, precipitation, masking, and many other types of operations may be involved in a method), and the analyst must pay attention

Table 5.10 Absorbed and Observed Colors

Color absorbed	Color observed
Red	Green
Orange	Blue-green
Yellow	Violet
Blue-green	Red
Blue	Orange
Violet	Yellow

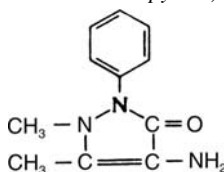
Table 5.11 Typical Reagents Used in Spectrophotometry

Aluminon (also called ammonium aurintricarboxylate)



This compound reacts with aluminum in a slightly acid solution (pH 4–5) to form an intense red color in solution. It detects 0.04–0.4 $\mu\text{g/mL}$ (ppm) of Al. Other elements, such as Be, Cr, Fe, Zr, and Ti, also react with aluminon. These elements must be removed if a sample is being analyzed for Al. The absorbance of the red solution is measured at 525 nm. The red color is the result of formation of a metal–dye complex called a “lake”.

4-Aminophenazone (also called 4-aminoantipyrine)

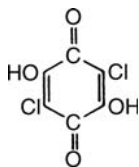


This compound reacts with a variety of phenols to give intensively colored compounds and will detect 0.02–6.4 ppm of phenol in water. Drinking water is steam-distilled to separate the volatile phenols from interfering compounds. The distillate is treated with the reagent, and the colored complex is extracted into CHCl_3 . The absorbance of the chloroform solution is measured at 460 nm. The reagent does not react with some para-substituted phenols, such as paracresol. This reaction is an example of the determination of organic compounds by spectrophotometric analysis following reaction with a color-producing reagent.

Thiourea
 H_2NCSNH_2

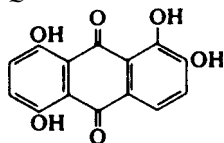
Thiourea will react with osmium, a very toxic element, in sulfuric acid solution to form a colored product. The absorbance is measured at 460 nm with a detection range of 8–40 ppm Os. The only interferences are Pd and Ru. Compared with the other reagents in this table, the sensitivity of this reagent is low. Interestingly, under different analytical conditions (different acid, pH) thiourea reacts with Bi. The absorbance of this product is measured at 322 nm, and detects 0.06–0.6 ppm of Bi. A number of elements such as Ag, Cu, Hg, and Pb interfere. This is an example of a reagent that works under different chemical conditions to produce a low-sensitivity determination for one element and a high-sensitivity determination for another.

Chloranilic acid

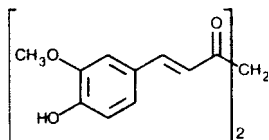


Chloranilic acid forms solutions that are intensely red. The addition of calcium to the solution precipitates the chloranilic acid and the intensity of the red diminishes. The change (loss) in color is a measure of the quantity of calcium added. Numerous other elements interfere with the procedure. This is an example of spectrophotometric analysis by loss of color after addition of the sample.

(continued)

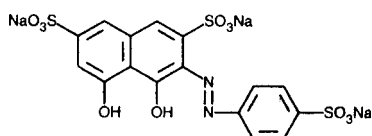
Table 5.11 Continued*Quinalizarin*

This reagent gives intensely colored solutions in aqueous solutions. In 93% w/w $\text{H}_2\text{SO}_4/\text{H}_2\text{O}$, the color is red. The presence of borate causes the color to become blue. Numerous other ions, such as Mg^{2+} , Al^{3+} , and Be^{3+} , also react with quinalizarin. This is an example of a change of color of the reagent after reaction with the sample.

Curcumin

Curcumin is a sensitive reagent for boron, detecting 0.01–0.1 ppm of B by absorbance at 555 nm. Fluoride, nitrate and nitrite interfere, but can be eliminated by separating the boron from the sample by distillation of B as a methyl borate ester.

SPADNS [also known as 4,5-dihydroxy-3-(2-hydroxy-5-sulphophenylazo)-2,7-naphthalenedisulfonic acid]



SPADNS (pronounced “spa-dens”) is used to determine fluoride ion in drinking water. The SPADNS dye reacts with zirconium to form a dark red Zr–dye “lake”. The F^- ion reacts to dissociate the Zr–dye complex and form $(\text{ZrF}_6)^{2-}$, which is colorless. The color of the solution decreases with increasing fluoride ion concentration. The absorbance is measured at 570 nm, with a range of 0.2–1.40 ppm F^- . There are both positive and negative interferences from chlorine, chloride, phosphate, sulfate, and other species in drinking water. This is another example of a reaction where the color is “bleached” with increasing concentration of analyte.

Source: Examples were extracted from Standard Methods for the Examination of Water and Wastewater, and the references by Dean and Dulski.

to all the details to achieve accurate and precise results. There is both science and art involved in performing many of these analyses. Many standard or regulatory methods (e.g., from Standard Methods for the Examination of Water and Wastewater, ASTM, EPA, etc.) have published precision and accuracy data in the methods. These are the precisions and accuracies that can be achieved by an experienced analyst.

5.5.3. Multicomponent Determinations

It has been seen that UV/VIS absorption peaks are generally broad, so if there are two compounds, X and Y, in solution, it is likely that they will not be completely resolved from each other. That is, both X and Y contribute to the absorbance at most wavelengths. It is possible to calculate the concentrations of X and Y from a series of measurements. Measurements must be made at a number of wavelengths equal to the number of components in the mixture. In this case, there are two components, so two wavelengths are needed.

Four calibration curves need to be prepared: X at λ_1 , X at λ_2 , Y at λ_1 , and Y at λ_2 . All calibration curves should be blank corrected to pass through the origin. The absorbance of the sample mixture is measured at λ_1 and at λ_2 .

Two equations can be written:

$$\begin{aligned}A_1 &= C_X S_{X1} + C_Y S_{Y1} \\A_2 &= C_X S_{X2} + C_Y S_{Y2}\end{aligned}\tag{5.3}$$

where A_1 is the absorbance of the unknown at λ_1 ; A_2 , the absorbance of the unknown at λ_2 ; C_X , the concentration of X in the unknown; C_Y , the concentration of Y in the unknown; S_{X1} , the slope of the calibration curve for X at λ_1 ; S_{X2} , the slope of the calibration curve for X at λ_2 ; S_{Y1} , the slope of the calibration curve for Y at λ_1 ; and S_{Y2} , the slope of the calibration curve for Y at λ_2 .

The absorbances and slopes are known; this leaves us with two equations and two unknowns, C_X and C_Y . The equations can be solved for the concentrations of X and Y in the unknown mixture. Dulski (see bibliography) gives an example of this approach with a method for the simultaneous determination of niobium and titanium by reaction with hydroquinone and measurement at 400 and 500 nm.

This same approach can be used for a mixture of three components. More complex mixtures can be unraveled through computer software that uses an iterative process at multiple wavelengths to calculate the concentrations. Mathematical approaches used include partial least squares, multiple least squares, principle component regression, and other statistical methods. Multicomponent analysis using UV absorption has been used to determine how many and what type of aromatic amino acids are present in a protein and to quantify five different hemoglobins in blood.

5.5.4. Other Applications

5.5.4.1. Reaction Kinetics

In common with other spectroscopic techniques, UV spectroscopy can be used to measure the kinetics of chemical reactions, including biochemical reactions catalyzed by enzymes. For example, suppose that two compounds A and B react to form a third compound C. If the third compound absorbs UV radiation, its concentration can be measured continuously. The original concentrations of A and B can be measured at the start of the experiment. By measuring the concentration of C at different time intervals, the kinetics of the reaction $A + B \rightarrow C$ can be calculated. Enzyme reactions are important biochemically and also analytically; an enzyme is very selective, even specific, for a given compound. The compound with which the enzyme reacts is called the substrate. If the enzyme assay is correctly designed, any change in absorbance of the sample will result only from reaction of the substrate with the enzyme. The rate of an enzyme reaction depends on temperature, pH, enzyme concentration and activity, and substrate concentration. If conditions are selected such that all of the substrate is converted to product in a short period of time, the amount of substrate can be calculated from the difference between the initial absorbance of the solution and the final absorbance. Alternatively, the other experimental variables can be controlled so that the rate of the enzyme reaction is directly proportional to substrate concentration.

5.5.4.2. Spectrophotometric Titrations

Many titration procedures in volumetric analysis use an indicator that changes color to signal the endpoint of the titration. For example, acid–base titrations are often performed

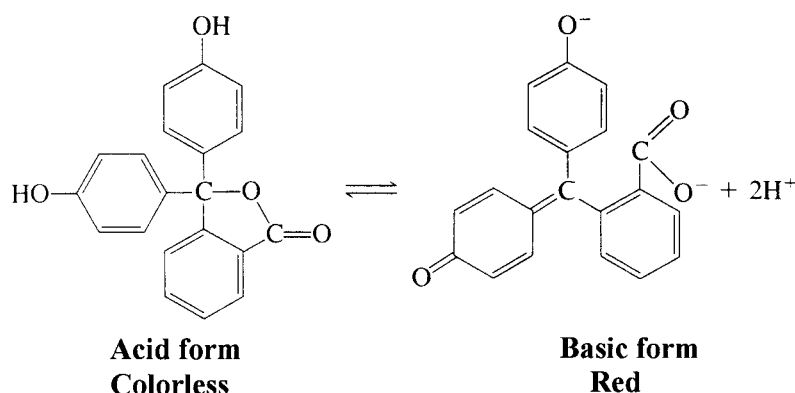


Figure 5.33 The structure of phenolphthalein in acidic solution and basic solution. The change in structure results in a change in the light absorbed by the molecule.

with indicators such as phenolphthalein. Figure 5.33 shows the structure of phenolphthalein in an acid solution and in a basic solution. As can be seen, the loss of protons results in a change in the structure of the molecule. As we know, this should result in a change in the energy levels in the molecule. In phenolphthalein, the energy level difference gives rise to the absorption of visible radiation when it is in an alkaline solution, but not in an acid solution. Phenolphthalein appears red in basic solution but colorless in acidic solution. Such structure changes and energy level changes are the basis of many acid–base indicators. Use of the human eye to detect the color change at the end of a titration is subject to the problems described at the beginning of the chapter. Each analyst may “see” the endpoint slightly differently from other analysts, leading to poor precision and possible errors. The use of a spectrophotometer to detect the color change is more accurate and reproducible. Use of the spectrophotometer also permits any change in absorbance in the UV or visible region by the titrant, analyte, or product to be used to determine the endpoint of the titration, so the method is not limited to reactions that use a colored indicator.

Spectrophotometric titrations have been used for redox titrations, acid–base titrations, and complexation titrations. The spectrophotometer can be used in a light scattering mode to measure the endpoint for a precipitation titration by turbidimetry. Spectrophotometric titrations can be easily automated.

5.5.4.3. Spectroelectrochemistry

Oxidation–reduction reactions of inorganic and organic compounds can be studied by using a combination of electrochemistry (Chapter 15) and spectroscopy. Diode array systems are usefully employed when transparent thin electrodes are used to study these reaction mechanisms. By taking the absorption spectra in rapid succession and accumulating the data, it is possible to detect and measure intermediates formed in complex reactions. This is much more reliable than using absorption at a single wavelength to measure the reactions, since the choice of the single wavelength is often made with the assumption that the intermediates and end products are well known and suitable absorption wavelengths are therefore easily chosen. This is often not the case. Using the diode array system, the complete UV absorption spectra can be obtained, and much more information on the identity and concentration of species is therefore available.

5.6. ACCURACY AND PRECISION IN UV/VIS ABSORPTION SPECTROMETRY

There are three major factors that affect the accuracy and precision of quantitative absorption measurements: the instrument, the skill of the analyst, and the method variables. Instruments vary in the quality of their optical, mechanical and electrical systems and also in their data processing. Each instrument has fixed limitations; these must be understood by the analyst and optimized when possible. Wavelength calibration must be checked routinely using recognized wavelength standards. Holmium oxide standards are commonly used for this purpose. Stray light, transmittance, resolution, and other instrument parameters should be checked regularly. The analyst must optimize slit widths if the instrument is equipped with variable slits; too narrow a slit width may result in errors due to a low signal-to-noise ratio, while too wide a slit width causes both loss of resolution and negative deviations from Beer's Law. Sample cells are very often the cause of error; they must be cleaned and handled properly for the best accuracy. Method variables include the quality of the reagents used, pH, temperature control, color stability, reaction kinetics, and stoichiometry. It may be necessary to remove interferences, to buffer the sample, to control exposure to air and light, and perform other chemical manipulations to achieve accurate results. The analyst must be trained to operate the instrument and to perform all the chemical manipulations required. Attention to detail, accurate recordkeeping, routine use of replicates, spiked samples, or reference materials, and the preparation and measurement of appropriate blanks and standards are the analyst's responsibility. The accuracy of a method will depend on the analyst, the method specificity, the removal or control of interferences, and finally on the spectrophotometer itself.

Spectrophotometric analyses are capable of being performed with relative standard deviations as low as 0.5%. Detection limits depend on the molar absorptivity of the transition being measured, but are often 0.05 ppm or lower for many analytes. The linear working range for spectrophotometry is generally only one to two orders of magnitude. This is a short linear range compared to fluorescence, as will be seen.

5.7. NEPHELOMETRY AND TURBIDIMETRY

Much of the theory and equipment used in spectrophotometry applies with little modification to **nephelometry** and **turbidimetry**. These fields involve the **scattering** of light by nontransparent particles suspended in a liquid; examples of such particles include fine precipitates and colloidal suspensions. In *nephelometry* we measure the amount of radiation *scattered* by the particles; in *turbidimetry* we measure the amount of light *not scattered* by the particles. These processes are illustrated in Figs. 5.34 and 5.35. The applications of nephelometry include the estimation of the clarity of drinking water, beverages, liquid pharmaceuticals, and other products where the transparency is important and in the determination of species that can be precipitated, such as calcium or barium by precipitation as the phosphate or sulfate insoluble salt. The quantity of calcium or barium present is measured by the amount of radiation scattered by the precipitated compound. From the intensity of scattered radiation, the original concentration of calcium or barium can be determined. Conversely, sulfate and phosphate can be determined by precipitation as the barium compound. Process analyzers using nephelometry or turbidimetry can be used to monitor the clarity of a plant stream or water treatment facility stream on a continuous basis.

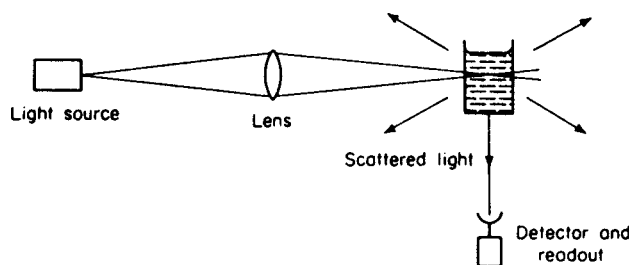


Figure 5.34 Schematic optical system for nephelometry.

When using nephelometry or turbidimetry for quantitative analysis, standard suspensions or standard turbid solutions are required for calibration. The precipitate or suspension standards must be prepared under rigidly controlled conditions. This is essential because the scattering of light depends on the *size*, *shape*, and *refractive index* of the *particles* involved, as well as on the concentration of particles. Some particles also absorb light, which will cause an error in the turbidity measurement. It is necessary for a given solution to produce the same number of particles of the same size and other properties listed for the degree of light scattering to be meaningful. Interferences include dirty sample cells, and any colored or absorbing species that will remove light from the light path. Any absorbance of light will result in an erroneously high turbidity, just as turbidity results in an erroneously high absorbance.

The wavelength of the light scattered most efficiently depends on the physical size of the scattering particles. From this, it can be reasoned that the size of the scattering particle may be determined if the wavelength of scattered light is accurately known. This type of light scattering forms the basis for the measurement of polymer molecular weights from the size of polymer molecules.

For water analysis, the formulation of turbid standards is very difficult, so most water laboratories use a synthetic polymer suspension as a standard. The formazin polymer suspension is easy to make and more stable and reproducible than adding clay or other particles to water to prepare standards. Alternatively, suspensions of polymer beads of the appropriate size can be used as scattering standards. (See Standard Methods for the Examination of Water and Wastewater for details.)

In the determination of a given species by a precipitation reaction, it is critical to control the experimental conditions. Two identical samples of equal concentration of analyte will scatter light equally only if they form the same number and size distribution of particles when they are precipitated. This depends on many experimental conditions, including the sample temperature, the rate at which the precipitant and the sample are

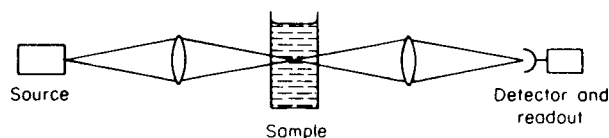


Figure 5.35 Schematic optical system for turbidimetry.

mixed, the degree of agitation or stirring, and the length of time the precipitates are allowed to stand before measurement. Procedures usually call for the use of a stopwatch to make all measurements at same point in time, such as 60 s after the reagent was added. Interferences include other particles, and colored or absorbing species. Sulfate in drinking water can be determined turbidimetrically by precipitation as barium sulfate over the range of 1–40 mg/L sulfate, with precision of about 2% RSD and accuracy, estimated by recovery of spiked samples, of about 90%.

5.8. MOLECULAR EMISSION SPECTROMETRY

5.8.1. Fluorescence and Phosphorescence

If a “black light” (UV light) is shone onto certain paints or certain minerals in the dark, they give off visible light. These paints and minerals are said to *fluoresce*. An energy diagram of this phenomenon is shown in Fig. 5.36. For fluorescence to occur, a molecule must absorb a photon and be promoted from its ground state to an excited vibrational state in a higher electronic state. There are actually *two* possible electronic transitions. Electrons possess the property of spin; we can think of this simplistically as the electron rotating either clockwise or counterclockwise. For two electrons to occupy the same orbital, their spins must be opposite to each other; we say that the spins are paired. If one electron is raised to the excited level without changing its spin, the electron in the excited level is still opposite in spin to the electron left behind in the valence level. This excited state of the molecule in which electron spins are paired is called a *singlet* state. If the electron spins are parallel (both spinning in the same direction as a result of the excited electron reversing its spin), the excited state is called a *triplet* state. Each “excited state” has both a singlet and corresponding triplet state. Singlet state energy levels are higher than the corresponding triplet state energies. Singlet states are designated S_1 , S_2 , S_3 , and so on; triplet states are designated T_1 , T_2 , T_3 , and so on. The ground state is a singlet state, S_0 . Figure 5.36 shows a ground state with the first excited singlet and triplet states. Some vibrational sub-levels of the excited states are also shown.

The molecule absorbs energy and an electron is promoted to one of the higher vibrational levels in the singlet state; this is a vibrationally excited electronic state. The vibrationally excited molecule will rapidly “relax” to the lowest vibrational level of the

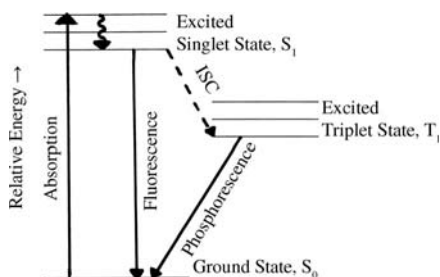


Figure 5.36 Schematic diagram of the ground state, excited singlet state, and excited triplet state, of a molecule. The wavy line denotes a radiationless transition from a higher vibrational level in the excited singlet state to the lowest vibrational level in the excited singlet state. The dotted arrow marked ISC shows the radiationless intersystem crossing from the excited singlet state to the excited triplet state. The length of the solid arrows denotes the relative energy of the transitions: absorption > fluorescence > phosphorescence. This results in $\lambda_{\text{phosphorescence}} > \lambda_{\text{fluorescence}} > \lambda_{\text{absorption}}$.

electronic excited state S_1 . This relaxation or loss of energy is a radiationless process, shown by the wavy arrow. Energy decreases but no light is emitted. Now the molecule can return to the ground state by emitting a photon equal to the energy difference between the two levels. This is the process of *fluorescence*: excitation by photon absorption to a vibrationally excited state, followed by a rapid transition between two levels with the same spin state (singlet to singlet, in this case) that results in the emission of a photon. The emitted photon is of lower energy (longer wavelength) than the absorbed photon. The wavelength difference is due to the radiationless loss of vibrational energy, depicted by the wavy line in Fig. 5.36. This type of fluorescence, emission of a longer wavelength than was absorbed, is what is usually seen in solutions; it is called *Stokes fluorescence*. The lifetime of the excited state is very short, on the order of 1–20 ns, so fluorescence is a virtually instantaneous emission of light following excitation. However, the lifetime of the fluorescent state is long enough that time-resolved spectra can be obtained with modern instrumentation. A molecule that exhibits fluorescence is called a *fluorophore*.

The transition from the singlet ground state to a triplet state is a forbidden transition. However, an excited singlet state can undergo a radiationless transition to the triplet state by reversing the spin of the excited electron. This is an energetically favorable process since the triplet state is at a lower energy level than the singlet state. This radiationless transition, shown schematically in Fig. 5.36 and 5.37, is called *intersystem crossing (ISC)*. The molecule can relax to the ground state from the triplet state by emission of a photon. This is the process of *phosphorescence*: excitation by absorption of light to an excited singlet state, then an ISC to the triplet state, followed by emission of a photon due to a triplet–singlet transition. The photon associated with phosphorescence is even lower energy (longer wavelength) than the fluorescence photon, as seen from the relative energy levels in Figs. 5.36 and 5.37. Because the triplet–singlet transition is forbidden, the lifetime of the triplet excited state is long, up to 10 s in some cases. The sample will “glow” for some time after the excitation light source is removed. “Glow in the dark” paint is an example of phosphorescent material.

Fluorescence and phosphorescence are both types of luminescence. They are specifically types of *photoluminescence*, meaning that the excitation is achieved by absorption of light. There are other types of luminescence. If the excitation of a molecule and emission of light occurs as a result of chemical energy from a chemical reaction, the luminescence is called *chemiluminescence*. The light emitted by a firefly is an example of *bioluminescence*.

As shown in Fig. 5.37, there are other ways for molecules to return to the ground state. Excited molecules may collide with other molecules; it is possible for energy to

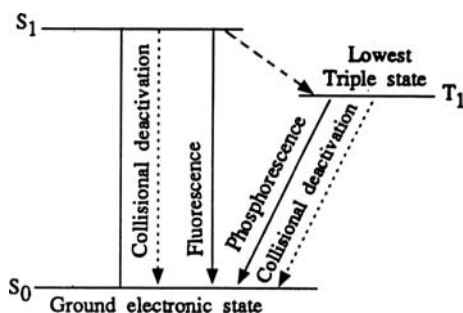


Figure 5.37 Processes by which an excited molecule can relax to the ground state. (Adapted from Guilbault, used with permission.)

be transferred during this collision. The molecule returns to the ground state but does not emit radiation. This is called collisional deactivation or *quenching*. Quenching occurs in solution by collision of the excited analyte molecule with the more numerous solvent molecules. Quenching in fluorescence is often a serious problem, but with care it can be minimized. Quenching by collision with the solvent molecules can be reduced by decreasing the temperature, thus reducing the number of collisions per unit time. The same result can be achieved by increasing the viscosity—for example, by adding glycerine. Dissolved oxygen is a strong quenching agent. It can be removed by bubbling nitrogen through the sample. Phosphorescence is very susceptible to quenching; the molecule in a triplet state has an extended lifetime in the excited state, so it is quite likely that it will collide with some other molecule and lose its energy of excitation without emitting a photon. Phosphorescence is almost never seen in solution at room temperature because of collisional deactivation. Low temperatures must be used and the analyte must be constrained from collision. This can be done for fluorescence and phosphorescence by converting the sample into a gel (highly viscous state), glass, or by adsorption of the analyte onto a solid substrate. “Organized” solvents such as surfactant micelles have been used successfully to observe room temperature phosphorescence and to greatly enhance fluorescence by reducing or eliminating collisional deactivation. Even with the appropriate experimental care, only a small fraction of available analyte molecules will actually fluoresce or phosphoresce, since radiationless transitions are very probable.

5.8.2. Relationship Between Fluorescence Intensity and Concentration

The intensity of fluorescence F is proportional to the amount of light absorbed by the analyte molecule. We know from Beer's Law that

$$\frac{I_1}{I_0} = e^{-abc} \quad (5.4)$$

so, subtracting each side of the equation from 1 gives:

$$1 - \frac{I_1}{I_0} = 1 - e^{-abc} \quad (5.5)$$

We multiply each side by I_0 :

$$I_0 - I_1 = I_0(1 - e^{-abc}) \quad (5.6)$$

Since

$$I_0 - I_1 = \text{amount of light absorbed}$$

the fluorescence intensity, F , may be defined as

$$F = (I_0 - I_1)\Phi \quad (5.7)$$

where Φ is the quantum efficiency or quantum yield. The quantum yield, Φ , is the fraction of excited molecules that relax to the ground state by fluorescence. The higher the value of Φ , the higher the fluorescence intensity observed from a molecule. A nonfluorescent molecule has $\Phi = 0$.

Therefore, fluorescence intensity is equal to:

$$F = I_0(1 - e^{-abc})\Phi \quad (5.8)$$

From Eq. (5.8), it can be seen that fluorescence intensity is related to the concentration of the analyte, the quantum efficiency, the intensity of the incident (source) radiation, and the absorptivity of the analyte. Φ is a property of the molecule, as is the absorptivity, a . A table of typical values of Φ for fluorescent molecules is given in Table 5.12. The absorptivity of the compound is related to the fluorescence intensity [Eq. (5.8)]. Molecules like saturated hydrocarbons that do not absorb in the UV/VIS region do not fluoresce.

The fluorescence intensity is directly proportional to the intensity of the source radiation, I_0 . In theory, the fluorescence intensity will increase as the light source intensity increases, so very intense light sources such as lasers, mercury arc lamps, or xenon arc lamps are frequently used. There is a practical limit to the intensity of the source because some organic molecules are susceptible to photodecomposition.

When the term abc is <0.05 , which can be achieved at low concentrations of analyte, the fluorescence intensity can be expressed as:

$$F = I_0 abc \Phi \quad (5.9)$$

That is, F , total fluorescence, $= kI_0c$, where k is a proportionality constant. At low concentrations, a plot of F vs. concentration should be linear. But only a portion of the total fluorescence is monitored or measured; therefore,

$$F' = Fk' \quad (5.10)$$

where F' is the measured fluorescence
and

$$F' = k'I_0c \quad (5.11)$$

where k' is another proportionality constant.

A plot of F vs. c is shown in Fig. 5.38. It is linear at low concentrations. The linear working range for fluorescence is about five orders of magnitude, from 10^{-9} to 10^{-4} M.

At higher concentrations the relationship between F and c deviates from linearity. The plot of F vs. c rolls over as seen in Fig. 5.39. It can be seen that at higher concentrations the fluorescence intensity actually decreases because the molecules in the outer part of the sample absorb the fluorescence generated by those in the inner part of the

Table 5.12 Fluorescence Quantum Yields, Φ

Compound	Solvent	Φ
9-Aminoacridine	Ethanol	0.99
Anthracene	Hexane	0.33
9,10-Dichloroanthracene	Hexane	0.54
Fluorene	Ethanol	0.53
Fluorescein	0.1 N NaOH	0.92
Naphthalene	Hexane	0.10
1-Dimethylaminonaphthalene-4-sulfonate	Water	0.48
Phenol	Water	0.22
Rhodamine B	Ethanol	0.97
Sodium salicylate	Water	0.28
Sodium sulfanilate	Water	0.07
Uranyl acetate	Water	0.04

Note: Solutions are 10^{-3} M, temperatures 21–25°C.

Source: Guilbault, used with permission.

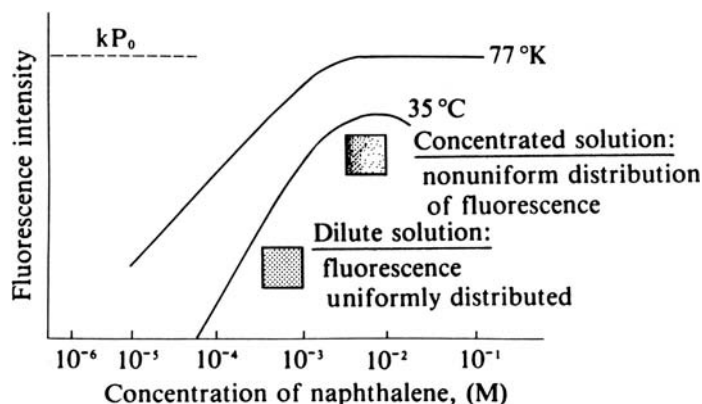


Figure 5.38 Dependence of fluorescence on the concentration of the fluorescing molecule. (From Guilbault, used with permission.)

sample. This is called the “inner cell” effect or self-quenching. In practice, it is necessary to recognize and correct for this effect. It is impossible to tell directly if the fluorescence measured corresponds to concentration A or concentration B as shown in Fig. 5.39. Both concentrations would give the same fluorescence intensity. Diluting the sample slightly can solve the dilemma. If the original concentration were A , then the fluorescence intensity would sharply decrease on dilution. On the other hand, if the concentration were B , then the fluorescence should increase on slight dilution of the sample.

5.9. INSTRUMENTATION FOR LUMINESCENCE MEASUREMENTS

A schematic diagram of a spectrofluorometer is shown in Fig. 5.40.

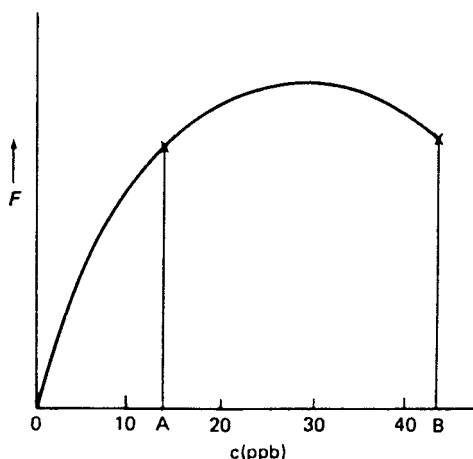


Figure 5.39 Fluorescence intensity at high concentrations of analyte. Note the reversal of fluorescence at high concentration. Concentrations A and B give the same fluorescence intensity and could not be distinguished from a single measurement.

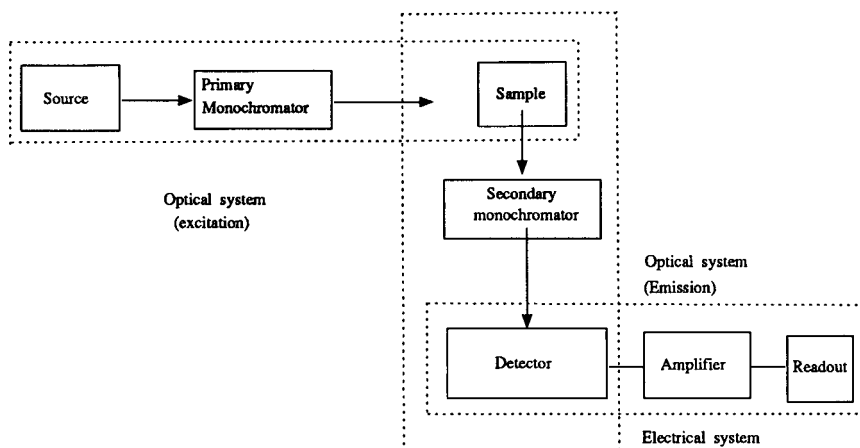


Figure 5.40 Block diagram of the optical components of a typical fluorometer. (From Guilbault, used with permission.)

5.9.1. Wavelength Selection Devices

Two monochromators are used, the primary or excitation monochromator, and the secondary or fluorescence monochromator. These are generally grating monochromators, although filters can be used for specific analyses. The excitation monochromator selects the desired narrow band of wavelengths that can be absorbed by the sample. The sample emits light in all directions. The second monochromator is placed at 90° to the incident light beam. The second monochromator is set to pass the fluorescence wavelength to the detector. The 90° orientation of the second monochromator is required to avoid the detector “seeing” the intense incident light, thus eliminating the background caused by the light source. Unlike absorption spectrophotometry, the measurement is not of the small difference between two signals, but of a signal with essentially no background. This is one reason for the high sensitivity and high linearity of fluorescence. Most fluorescence instruments are single beam instruments. This means that changes in the source intensity will result in changes in the fluorescence intensity. To compensate for changes in the source intensity, some instruments split off part of the source output, attenuate it, and send it to a second detector. The signals from the two detectors are used to correct for drift or fluctuations in the source.

The 90° geometry is the most common orientation for measuring fluorescence and works very well for solution samples that do not absorb strongly. Other angles are used in specific applications. For strongly absorbing solutions or for solid samples such as thin layer chromatography plates, fluorescence is measured from the same face of the sample illuminated by the source. This is called front-surface geometry. It is shown schematically for a solid sample in Fig. 5.41.

5.9.2. Radiation Sources

The fluorescence intensity is directly proportional to the intensity of the light source. Therefore intense sources are preferred. Excitation wavelengths are in the UV and visible regions of the spectrum, so some of the same sources used in UV/VIS absorption spectrometry are used for fluorescence. The optical materials will of course be the same—quartz for the UV, glass for the visible region.

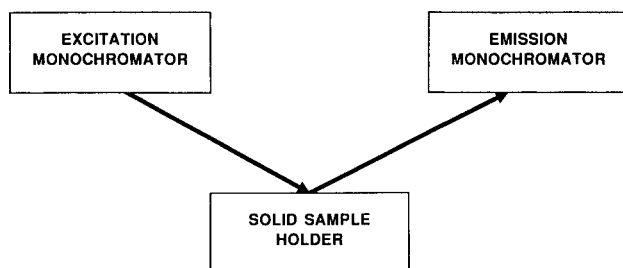


Figure 5.41 Front surface fluorescence geometry for a solid sample. (From Froelich and Guilbault, used with permission.)

Mercury or xenon arc lamps are used. A schematic of a xenon arc lamp is given in Fig. 5.42. The quartz envelope is filled with xenon gas, and an electrical discharge through the gas causes excitation and emission of light. This lamp emits a continuum from 200 nm into the IR. The emission spectrum of a xenon arc lamp is shown in Fig. 5.43. Mercury lamps under high pressure can be used to provide a continuum, but low-pressure Hg lamps, which emit a line spectrum, are often used with filter fluorometers. The spectrum of a low-pressure Hg lamp is presented in Fig. 5.44.

Because of their high intensity, laser light sources are ideal sources for fluorescence. The laser must exhibit a wide range of emission wavelengths, so tunable dye lasers have



Figure 5.42 Compact xenon arc lamp used in fluorometers. The quartz envelope is filled with xenon gas. The lamp is ignited by a 10–20 kV pulse across the electrodes. (From Froelich and Guilbault, used with permission.)

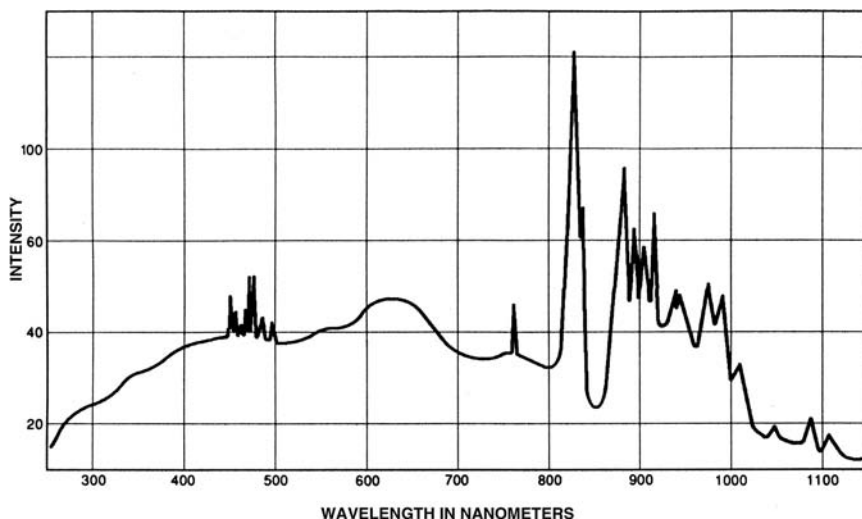


Figure 5.43 Spectral output of a compact xenon arc lamp. (From Froelich and Guilbault, used with permission.)

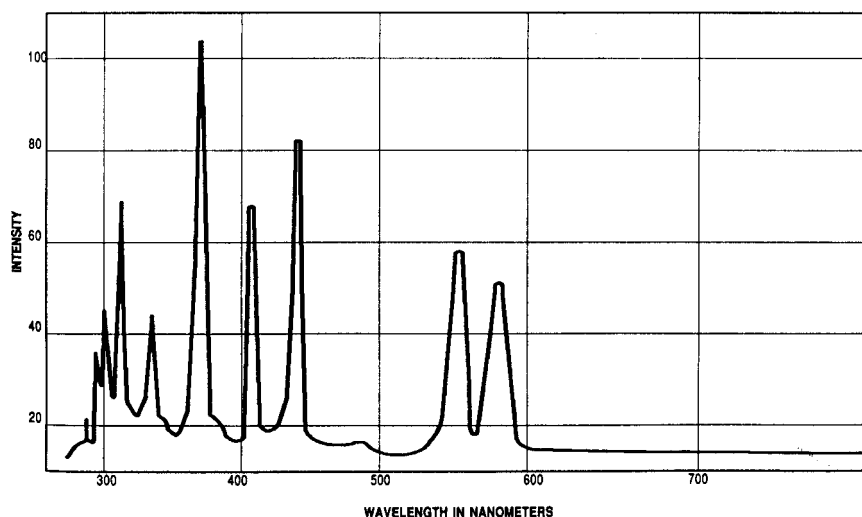


Figure 5.44 Spectral output of a mercury arc lamp, used as a source in fluorometers. (From Froelich and Guilbault, used with permission.)

been the only choice until recently. The dye lasers are generally pumped by an Nd:YAG laser. Nd stands for neodymium and YAG is yttrium aluminum garnet. These pumped dye laser systems are very expensive and complicated to operate. They have much greater intensity output than lamps and so enable lower detection limits to be achieved. Recent advances in solid-state lasers have made small, less expensive visible wavelength lasers available. Solid-state UV lasers are available, but do not have the required intensity for use as a fluorescence source.

5.9.3. Detectors

The most common detector in use is the PMT. The operation of the PMT was described earlier in this chapter. Because the signal is small due to the low concentrations of analyte used, the PMT is often cooled to subambient temperature to reduce noise. The limitation of the PMT is that it is a single wavelength detector. This requires that the spectrum be scanned. As we have discussed, scanning takes time and is not suitable for transient signals such as those from a chromatographic column. Diode array detectors are now used to collect the entire spectrum at once instead of scanning. The CCD, a 2D array detector, is another alternative to scanning in fluorescence spectrometry. Both PDA and CCD detectors can be used with LC or CE systems for separation and detection of fluorescent compounds or “tagged” compounds in mixtures. LC and CE are discussed in Chapter 13.

5.9.4. Sample Cells

The most common cell for solutions is a 1 cm rectangular quartz or glass cuvet with **four** optical windows. For extremely small volumes, fiber optic probes, microvolume cells, and flow cells are available. Gas cells, and special sample compartments for solid samples are commercially available.

5.10. ANALYTICAL APPLICATIONS OF LUMINESCENCE

Fluorescence occurs in molecules that have low energy $\pi > \pi^*$ transitions; such molecules are primarily aromatic hydrocarbons and polycyclic aromatic compounds. Examples include those in Table 5.12, as well as compounds like indole and quinoline. Molecules with rigid structures exhibit fluorescence; the rigidity evidently decreases the probability of a radiationless deactivation. Some organic molecules increase their fluorescence intensity on complexation with a metal ion. The resulting complex structure is more rigid than the isolated organic molecule in solution. Molecules that fluoresce can be measured directly; the number of such molecules is estimated to be between 2000 and 3000 from the published literature. There are several compounds that exhibit strong fluorescence; these can be used to derivatize, complex or “tag” nonfluorescent species, thereby extending the range of fluorescence measurements considerably. Other analytes are very efficient at quenching the fluorescence of a fluorophore; there are quantitative methods based on fluorescence quenching.

A fluorometric analysis results in the collection of two spectra, the excitation spectrum and the emission spectrum. The excitation spectrum should be the same as the absorption spectrum obtained spectrophotometrically. Differences may be seen due to instrumental factors, but these are normally small, as seen in Fig. 5.45, which shows the absorption and excitation spectra for Alizarin garnet R, a fluorometric reagent for aluminum ion and fluoride ion. The longest wavelength absorption maximum in the excitation spectrum is chosen as the excitation wavelength; this is where the first monochromator is set to excite the sample. It would seem reasonable to choose the wavelength that

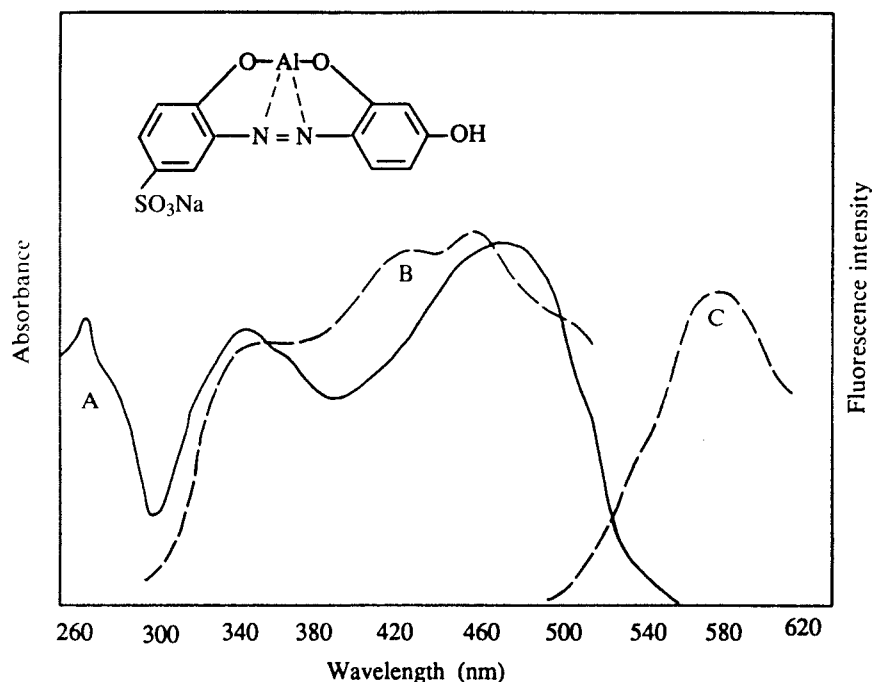


Figure 5.45 Absorption and fluorescence spectra of the aluminum complex with acid Alizarin garnet R (0.008%): Curve A, the absorption spectrum; Curve B, the fluorescence excitation spectrum; Curve C, the fluorescence emission spectrum. (From Guilbault, used with permission.)

provides the most intense fluorescence as the excitation wavelength, but often short wavelengths from the high intensity sources used can cause a compound to decompose. The emission spectrum is collected by the second monochromator. The emission or fluorescence spectrum for Alizarin garnet R is shown in Fig. 5.45. Similar excitation and emission spectra are shown in Fig. 5.46 for quinine and anthracene. Note, especially for anthracene, that the fluorescence (emission) spectrum is almost a mirror image of the excitation spectrum. The shape of the emission spectrum and wavelength of the fluorescence maximum do not depend on the excitation wavelength. The same fluorescence spectrum is obtained for any wavelength the compound can absorb. However, the intensity of the fluorescence is a function of the excitation wavelength.

Fluorometry is used in the analysis of clinical samples, pharmaceuticals, natural products, and environmental samples. There are fluorescence methods for steroids, lipids, proteins, amino acids, enzymes, drugs, inorganic electrolytes, chlorophylls, natural and synthetic pigments, vitamins, and many other types of analytes. The detection limits in fluorometry are very low. Detection limits of 10^{-9} M and lower can be obtained. Single molecule detection has been demonstrated under *extremely* well controlled conditions. This makes fluorometry one of the most sensitive analytical methods available. Therefore, the technique is widely used in quantitative trace analysis. For example, Table 5.10 indicated that Al^{3+} could be detected spectrophotometrically using Aluminon at about 0.04 ppm in solution and fluoride could be detected at 0.2 ppm with SPADNS. Using fluorometry and Alizarin garnet R, whose structure is shown in Fig. 5.47, Al^{3+} can be determined at 0.007 ppm and F^- at 0.001 ppm. The strongly fluorescent compounds like fluorescein can be detected at part per trillion levels (ng/mL in solution), so use of

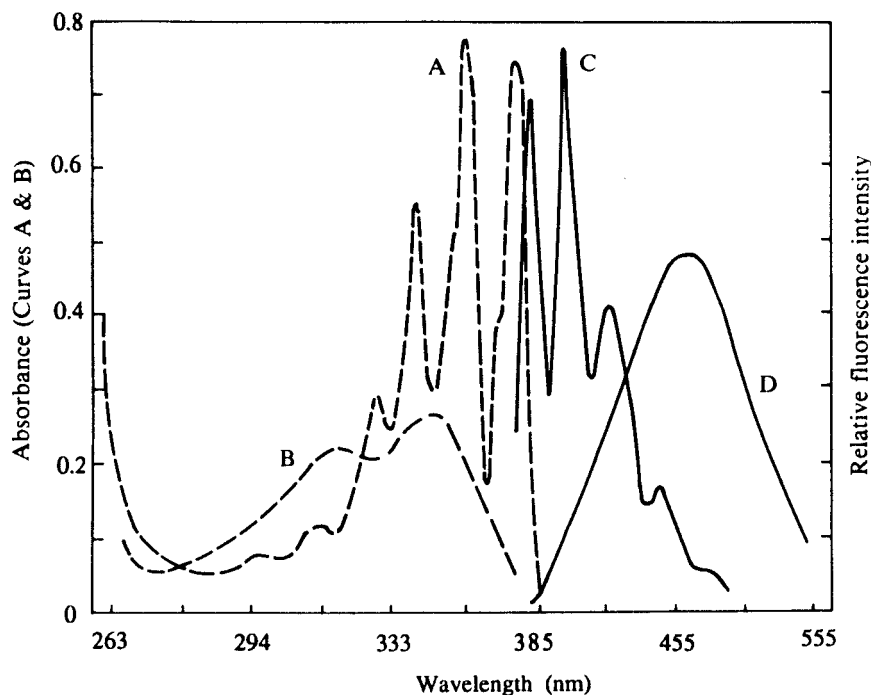


Figure 5.46 Absorption and fluorescence spectra of anthracene and quinine: Curve A, anthracene absorption; Curve B, quinine absorption; Curve C, anthracene fluorescence; Curve D, quinine fluorescence. (From Guilbault, used with permission.)

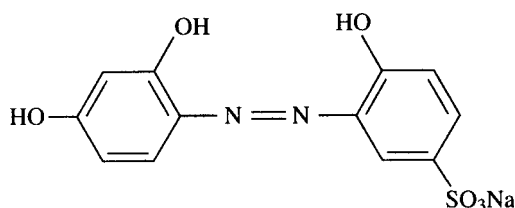


Figure 5.47 Structure of Alizarin garnet R.

such a compound as a “tag” can result in a very sensitive analytical method for many analytes.

5.10.1. Advantages of Fluorescence and Phosphorescence

The advantages of fluorescence and phosphorescence for analyses of molecules include extremely high sensitivity, high specificity, and a large linear dynamic range. The sensitivity is a result of the direct measurement of the fluorescence or phosphorescence signal against a zero background signal, as described. Specificity is a result of two factors: first, not all molecules fluoresce; therefore, many molecules are eliminated from consideration; and second, two wavelengths, excitation and emission, are used in fluorometry instead of one in spectrophotometry. It is not likely that two different compounds will emit at the same wavelength, even if they absorb the same wavelength and vice versa. If the fluorescing compounds have more than one excitation or fluorescent wavelength, the difference in either the emission spectrum or the excitation spectrum can be used to measure mixtures of compounds in the same solution. In Fig. 5.46, for example, the excitation spectra of quinine and anthracene overlap, but they do not emit at the same wavelengths, so the two compounds could be measured in a mixture. The linear dynamic range in fluorometry is six to seven orders of magnitude compared to one to two orders of magnitude that can be achieved in spectrophotometry.

5.10.2. Disadvantages of Fluorescence and Phosphorescence

Other compounds that fluoresce may need to be removed from the system if the spectra overlap. This can be done, for example, by column chromatography. Peaks may appear in the fluorescence spectrum that are due to other emission and scattering processes; Rayleigh, Tyndall, and Raman scattering may be seen because of the high intensity of the light source used. Peaks due to fluorescent impurities may occur.

Reversal of fluorescence intensity or self-quenching at high concentrations is a problem in quantitative analysis but can be eliminated by successive dilutions. Quenching by impurities can also occur and can cause significant problems in analysis. Changes in pH can frequently change structure, as we saw with phenolphthalein in Fig. 5.33, and thereby change fluorescence intensity; pH must therefore be controlled. Temperature and viscosity need to be controlled as well for reproducible results.

Photochemical decomposition or photochemical reaction may be induced by the intense light sources used. In general, the approach of using the longest excitation wavelength possible and the shortest measurement time possible will minimize this problem.

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SUGGESTED EXPERIMENTS

- 5.1 Add a drop of toluene to a UV absorption cell and cap or seal the cell. Record the absorption spectrum of toluene vapor over the UV range (220–280 nm) several times, varying the slit widths but keeping the scan speed constant. For example, slit widths of 0.1, 0.5, 1, and 5 nm can be used. Explain what happens to the spectral resolution as the slit width is changed.
- 5.2 Record the absorption spectrum of a solution of pure octane. Record the absorption spectrum of a 0.02% v/v toluene in octane solution from 220 to 280 nm. For a double-beam spectrometer, pure octane can be put into the reference cell. Compare with Experiment 5.1. Explain your observations. Change the slit width as in Experiment 5.1 and observe what happens to the resolution.
- 5.3 Record the absorption spectrum between 400 and 200 nm of (a) 1-octene, (b) 1,3-butadiene, and (c) a nonconjugated diolefin (e.g., 1,4-pentadiene). What is the effect of a conjugated system on the absorption spectrum? What does the spectrum tell you about the relative energy of the molecular orbitals in each compound?
- 5.4 Record the absorption spectrum of a polynuclear aromatic compound such as anthracene and of a quinonoid such as benzoquinone. How does the structure of the compound affect the spectrum?
- 5.5 From Experiment 5.2, choose a suitable absorbance wavelength (or wavelengths) for toluene. Based on the maximum absorbance for your 0.02% solution, prepare a series of toluene in octane solutions of higher and lower concentrations. (For example, 0.1%, 0.05%, 0.01% v/v toluene in octane might be suitable.) Measure the absorbance of each solution (using octane as the reference) at your chosen wavelength.
- 5.6 Using the absorbance data obtained from Experiment 5.5, plot the relationship between the absorbance A and concentration c of toluene at each wavelength chosen. Indicate the useful analytical range for each wavelength.
- 5.7 Prepare a standard solution of quinine by dissolving a suitable quantity of quinine in water. Record the UV absorption spectrum of the solution between 500 and 200 nm. Record the absorption spectra of several commercial brands of quinine water (after allowing the bubbling to subside). Which brand contained the most quinine? (Tonic water contains quinine.)
- 5.8 Prepare ammonium acetate buffer by dissolving 250 g of ammonium acetate in 150 mL of deionized water and then adding 700 mL of glacial acetic acid. Prepare a 1,10-phenanthroline solution by dissolving 100 mg 1,10-phenanthroline monohydrate in 100 mL deionized water to which two drops of conc. HCl have been added. 1 mL of this reagent will react with no more than 100 μg of ferrous ion, Fe^{2+} . Prepare a stock ferrous iron solution containing 1 g/L of ferrous sulfate. By taking aliquots of the stock solution and diluting, prepare four standard solutions and a blank in 100 mL volumetric flasks as follows: Pipet 0, 100, 200, 300, and 400 μg of Fe^{2+} into 100 mL flasks, then add 2 mL conc. HCl to each flask. Add 10 mL of ammonium acetate buffer solution and 4 mL of 1,10-phenanthroline solution. Dilute to the mark with deionized

water. Mix completely and allow to stand for 15 min for color development. Measure the absorbance at 510 nm. Correlate the absorbance with the concentration of iron in the solutions (remember to subtract the blank) and prepare a calibration curve. *Note:* all reagents used should be low in iron or trace metal grade. The use of this method for determining total iron, ferric iron, and ferrous iron in water may be found in Standard Methods for the Examination of Water and Wastewater. Sample preparation is required for real water samples, as the reagent only reacts with ferrous ion.

PROBLEMS

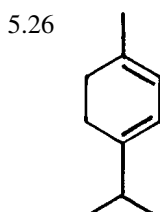
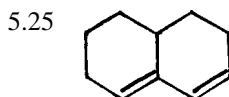
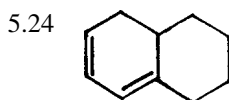
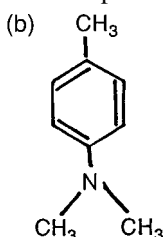
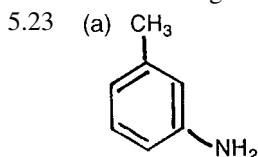
- 5.1 What types of molecules are excited by UV radiation? Why?
- 5.2 Indicate which of the following molecules absorb UV radiation and explain why: (a) heptane, (b) benzene, (c) 1,3-butadiene, (d) water, (e) 1-heptene, (f) 1-chlorohexane, (g) ethanol, (h) ammonia, and (i) *n*-butylamine.
- 5.3 Draw a schematic diagram of a double-beam spectrophotometer. Briefly explain the function of each major component.
- 5.4 List the principal light sources used in UV/VIS spectrometry.
- 5.5 Radiation with a wavelength of 640 nm is dispersed by a simple grating monochromator at an angle of 20° . What are the other wavelengths of radiation that are dispersed at the same angle by this grating (lowest wavelength 200 nm)?
- 5.6 Explain the operating principle of the photomultiplier tube.
- 5.7 What are the limitations of UV absorption spectroscopy as a tool for qualitative analysis?
- 5.8 (a) Plot a calibration curve for the determination of monochlorobenzene from the data listed below.
(b) Three samples of monochlorobenzene were brought in for analysis. The samples transmitted (1) 90%, (2) 85%, and (3) 80% of the light under the conditions of the calibration curve just prepared. What was the concentration of monochlorobenzene in each sample?

Concentration (ppm)	Absorbance
1.2	0.24
2.5	0.50
3.7	0.71
5.1	0.97
7.2	1.38
9.8	1.82

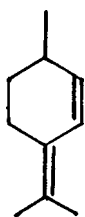
- 5.9 Several samples of monochlorobenzene were brought to the laboratory for analysis using the calibration curve in Problem 5.8. The absorbance of each sample is listed below.

Sample	Absorbance
A	0.400
B	0.685
C	0.120
D	0.160
E	3.0

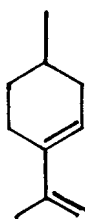
- (a) What are the respective concentrations of monochlorobenzene in samples A–D? (b) What is the problem with Sample E? How could the analysis of sample E be obtained?
- 5.10 Which of the following absorb in the UV region? (a) N_2 , (b) O_2 , (c) O_3 , (d) CO_2 , (e) CH_4 , (f) C_2H_4 , (g) I_2 , (h) Cl_2 , (i) Cyclohexane, and (j) C_3H_6 .
- 5.11 Why does phenolphthalein change color when going from an acid to a basic solution?
- 5.12 Why do UV absorption spectra appear as broad bands?
- 5.13 What causes the blue shift and the red shift in spectra?
- 5.14 Why do D_2 lamps emit a continuum and not line spectra?
- 5.15 How does a pn diode work?
- 5.16 Describe a diode array.
- 5.17 Describe the processes of UV molecular fluorescence and phosphorescence.
- 5.18 What is the relationship between fluorescence and excitation light intensity I_0 ?
- 5.19 Explain the reversal of fluorescence intensity with increase in analyte concentration. How is this source of error corrected?
- 5.20 Draw a schematic diagram of the instrumentation used for measuring UV fluorescence intensity.
- 5.21 (a) What interferences are encountered in UV fluorescence?
(b) Why is phosphorescence not used as extensively as fluorescence for analytical measurements?
- 5.22 From the emission spectra of quinine and anthracene (Fig. 5.46), pick a wavelength that will permit you to determine quinine in a mixture of quinine and anthracene. Do the same for anthracene. Can you use the excitation spectra to distinguish between the two compounds? Explain.
- Calculate the wavelength of the absorption maximum of the following compounds:



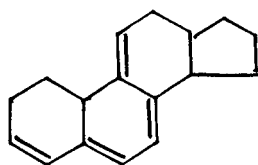
5.27



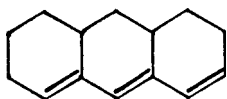
5.28



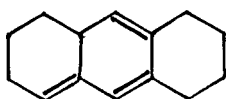
5.29



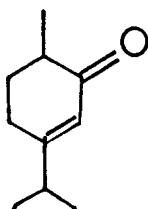
5.30



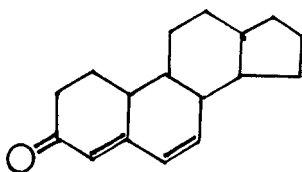
5.31



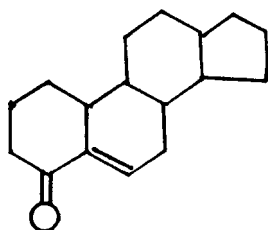
5.32



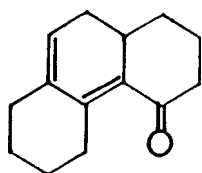
5.33



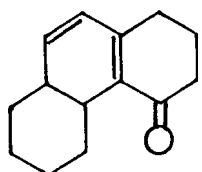
5.34



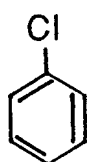
5.35



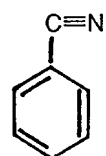
5.36



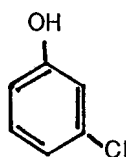
5.37



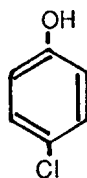
5.38



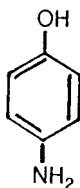
5.39



5.40



5.41



5.42 If a solution appears blue when a white light is passed through it, what colors has the solution absorbed?

5.43 State Beer's Law. What conditions must be met for Beer's Law to apply? Complete the following table:

Solution	Absorption (%)	T	A	Concentration (ppm)
1	1			1
2	13			6
3	30			15
4	55			34
5	80			69

5.44 Assuming the data obtained in Problem 5.43 were for a calibration curve and the same cell was used for all measurements, complete the following table:

Solution	Absorption (%)	T	Concentration (ppm)
A	30		
B	3		
C	10		
D	50		
E	70		

5.45 What is the relationship between the absorption cell length b and the absorbance A ? Complete the following table (the concentration c was equal in all cases):

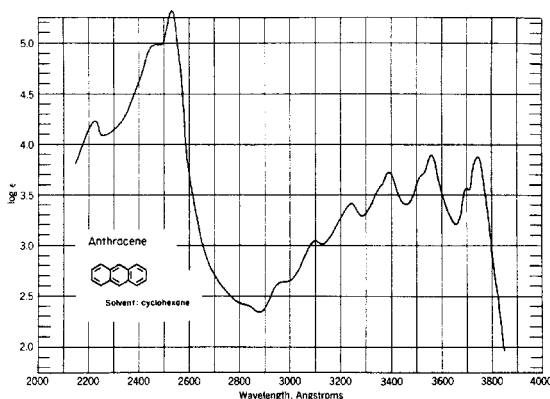
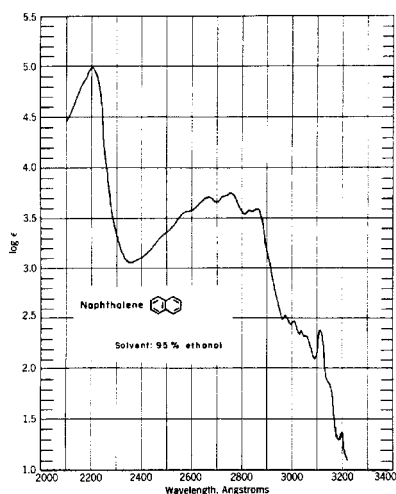
Sample	Path length b (cm)	A
1	0.1	0.01
2	0.5	
3	1.0	
4	2.0	
5	5.0	

5.46 Name three reagents used for quantitative UV/VIS spectrometric analysis and the elements they are used to determine.

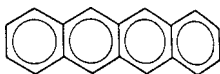
5.47 Name two fluorometric reagents. What are the structural characteristics that make a molecule fluoresce?

5.48 Below are the absorption spectra of naphthalene and anthracene (from Jaffé and Orchin, with permission). Their structures are shown on the spectra.

These molecules are polycyclic aromatic hydrocarbons, formed by fusing together benzene rings.

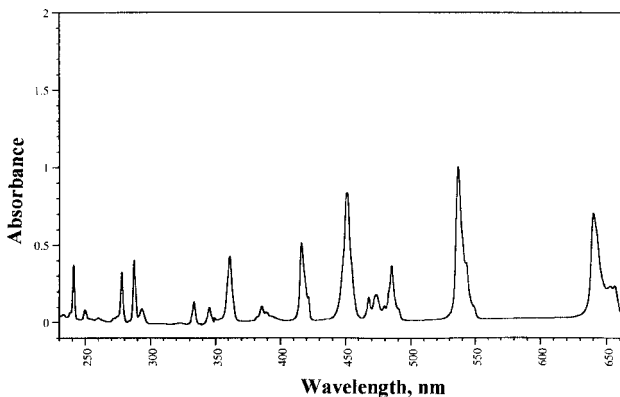


(a) Tabulate the wavelengths for the absorption maxima for these two compounds and in the spectrum of benzene (Fig. 5.12 in the text). What trend do you observe? (b) What transition is causing the peaks observed in these compounds? (c) Explain the trend you observe in the absorption maxima. (d) The next larger molecule in this family is naphthacene, a four-ring compound, with the structure shown here:



Predict where the absorption maxima will occur for naphthacene. Explain your prediction.

5.49 The UV/VIS absorption spectrum shown here is the spectrum of holmium oxide (from Starna Cells, Inc., www.starna.com, with permission). It is a rare earth oxide and is available in high purity.



(a) Qualitatively, what differences do you see between this spectrum and the spectrum of an organic molecule such as pyridine (Fig. 5.1)? Why do you think they are different in appearance? (b) Consider the spectrum. Think of how you might use holmium oxide to check on the operation of your UV/VIS spectrometer. What could you check?

6

Atomic Absorption Spectrometry

The basis of atomic absorption spectrometry (AAS) is the absorption of discrete wavelengths of light by ground state, gas phase free atoms. Free atoms in the gas phase are formed from the sample by an “atomizer” at high temperature. AAS was developed in the 1950s by Alan Walsh and rapidly became a widely used analytical tool. AAS is an **elemental analysis** technique capable of providing quantitative information on ~70 elements in almost any type of sample. As an elemental analysis technique, it has the significant advantage in many cases (but not all) of being practically independent of the chemical form of the element in the sample. A determination of cadmium in a water sample is a determination of the total cadmium concentration. It does not matter whether the cadmium exists as the chloride, sulfate, or nitrate, or even if it exists as a complex or an organometallic compound, if the proper analysis conditions are used. Concentrations as low as ppt levels of some elements in solution can be determined, and AAS is used routinely to determine ppb and ppm concentrations of most metal elements. Another principal advantage is that a given element can be determined in the presence of other elements, which do not interfere by absorption of the analyte wavelength. Therefore it is not necessary to separate the analyte from the rest of the sample (the matrix). This results in rapid analysis times and eliminates some sources of error. This is not to say that AAS measurements are completely free from interferences; both chemical and spectral interferences do occur and must be compensated for, as will be discussed. The major disadvantages of AAS are that no information is obtained on the chemical form of the analyte (no “speciation”) and that often only one element can be determined at a time. This last disadvantage makes AAS of very limited use for qualitative analysis. AAS is used almost exclusively for quantitative analysis of elements, hence the use of the term “spectrometry” in the name of the technique instead of “spectroscopy”.

6.1. ABSORPTION OF RADIANT ENERGY BY ATOMS

AAS is based on the absorption of radiant energy by free gas phase atoms. In the process of absorption, an atom changes from a low-energy state to a higher energy state as discussed in Chapter 2. Gas phase atoms do not vibrate in the same sense that molecules do. Also, they have virtually no rotational energy. Hence no vibrational or rotational energy is involved in the electronic excitation of atoms. As a result, atomic absorption spectra consist of a few very narrow absorption lines, in contrast to the wide bands of energy absorbed by molecules in solution.

Each element has a specific number of electrons “located” in an orbital structure that is unique to each element. The lowest energy electronic configuration of an atom is called